Affordable, Portable, Real-Time Chlorophyll Sensor

Physics Report | Group 04

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2026-02-18

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Executive Summary

In this project, we developed a low-cost, high-precision chlorophyll sensor that uses **Laser-Induced Fluorescence (LIF)** coupled with **tinyML** algorithms. Our goal was to build a tool suitable for real-world agricultural and environmental monitoring. We successfully prototyped a system that can measure both Chlorophyll-a (Chl-a) and Chlorophyll-b (Chl-b) concentrations in real-time, bringing lab-grade analysis into a portable, affordable format.

1 Introduction

1.1 Why Chlorophyll Matters

Knowing the chlorophyll content is one of the easiest ways to check if a plant is healthy. Whether it's for precision farming or monitoring algae in local water bodies, getting these numbers quickly is important.

1.2 Our Approach

If you want accurate chlorophyll measurements today, you usually have to send samples to a lab for expensive HPLC or Mass Spectrometry tests. Portable meters do exist, but they often only give relative "greenness" values rather than absolute concentrations.

- **The Problem**: Professional chlorophyll analysis is usually too expensive or too slow for field use.
- **What we did**: We built a prototype sensor that uses laser excitation and edge computing to give us absolute concentration readings right on the device.

2 Theoretical Framework

2.1 Physics of Laser-Induced Fluorescence (LIF)

The sensor employs a flow-through system where samples are excited by targeted wavelengths: * **Laser 405 nm**: Optimized for Chl-a excitation. * **Laser 450 nm**: Optimized for Chl-b excitation.

2.2 Spectral Red Shift & Concentration Features

A critical physical observation in this study is the **Red Shift** of the fluorescence peak as concentration increases. This phenomenon occurs due to: * **Inner Filter Effect (IFE)**: Re-absorption of the shorter-wavelength side of the emission spectrum by the chlorophyll matrix. * **Self-Absorption**: The overlap between chlorophyll absorption and emission bands causes selective re-absorption at shorter wavelengths. * **Concentration Quenching**: At high concentrations, molecular aggregation further shifts the emission to longer wavelengths.

These effects produce three measurable spectral features that vary systematically with concentration, as confirmed by our simulation study:

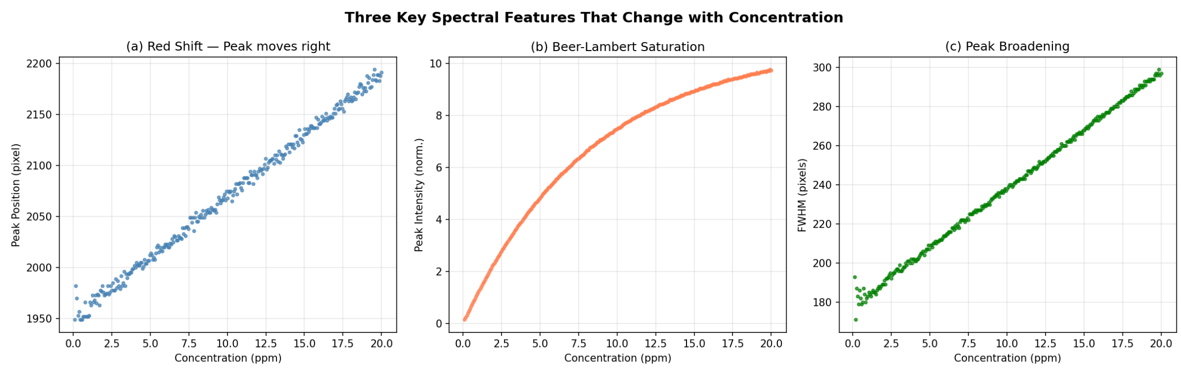


Figure 1: Three key spectral features that change with chlorophyll concentration: (a) peak position shifts ~240 pixels across the 0–20 ppm range, (b) intensity saturates following Beer-Lambert law, and (c) FWHM broadens linearly.

Unlike static peak-height models, our PLS and SVR models analyze the **entire spectral curve**, allowing them to correlate this non-linear shift with absolute concentration. Importantly, our simulation study confirmed that **retaining the red shift** (i.e., not aligning peaks) provides superior model performance compared to peak-aligned spectra.

3 Experimental Setup & Hardware

3.1 Our Hardware Setup

- **The Brain**: We used an STM32H743VIT6. It's a high-performance chip running at 480MHz, which gave us more than enough speed to handle the CCD data and run our machine learning models on the fly.
- **The Sensor**: For the "eyes" of our instrument, we used a Toshiba TCD1304 linear CCD. With 3648 pixels, it was perfect for resolving the small shifts in the fluorescence spectrum.
- **Light & Optics**:

  - **Excitation**: We used two lasers (405 nm and 450 nm). A small servo motor moves a beam combiner to switch between these two excitation sources.
  - **Splitting the Light**: We used a 600 lines/mm holographic grating to spread the fluorescence light across the CCD.
  - **Alignment**: We used a cylindrical lens and a convex mirror to guide the laser. One of our biggest challenges was the manual alignment of the objective lens—it's very sensitive and took a lot of patience to get right before each measurement session.
- **Custom PCB**: We designed our own board in **KiCad** and had it made by **JLCPCB**. This helped us keep the electronics compact and reduced the wiring noise.

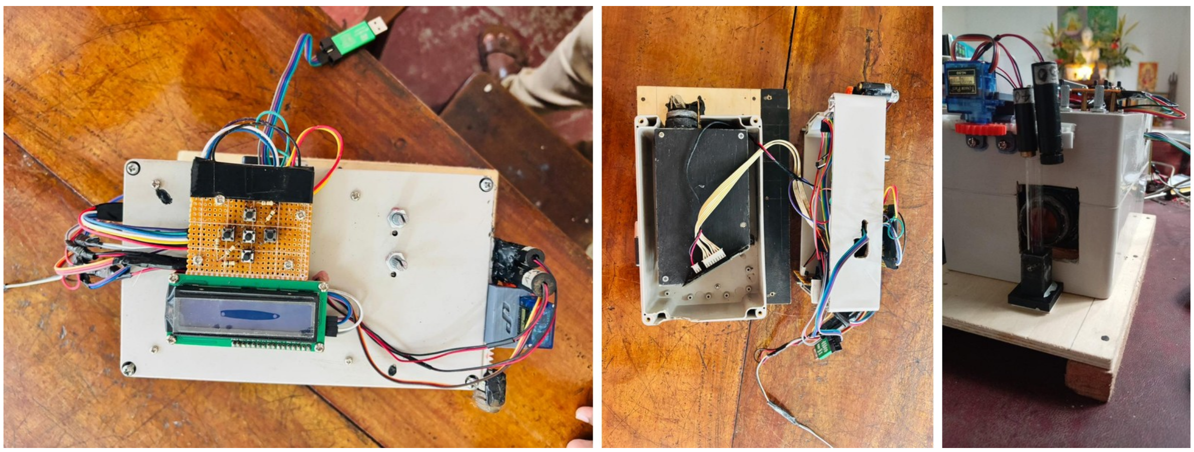


Figure 2: Instrument overview — (left) top-down view of the assembled optical bench, (centre) interior showing the diffraction grating and CCD mount, (right) front panel with cuvette holder and I/O connections.

3.2 System Connectivity

The following diagram illustrates the physical and logical connections between the core components:

3.3 Information Pipeline

The data flow from raw photons to on-device concentration prediction is summarized below:

3.4 Circuitry and PCB Design

The system's electronic backbone is a custom-designed PCB optimized for low-noise CCD readout and high-speed signal processing.

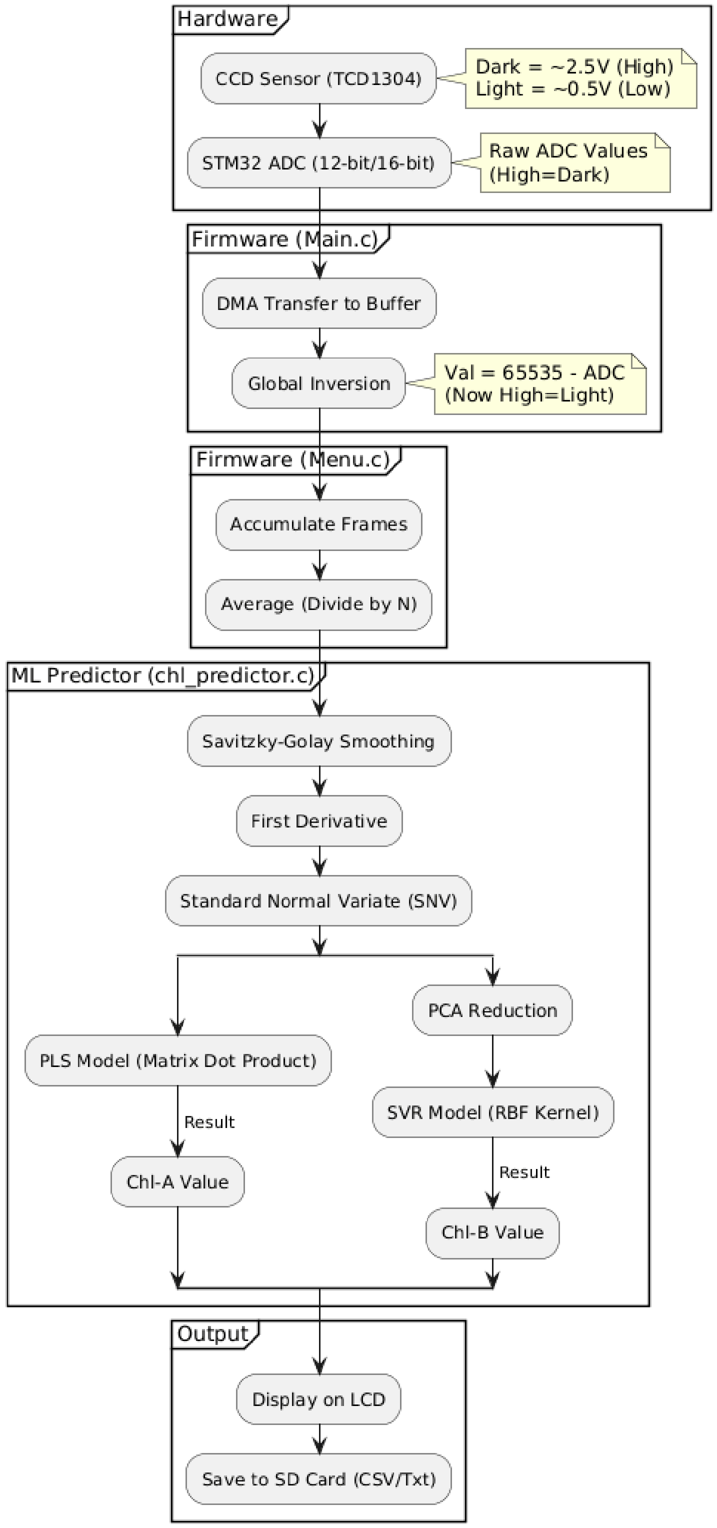


Figure 3: System connectivity diagram showing physical and logical connections between STM32H743VIT6, TCD1304 CCD, lasers, servo, LCD, SD card, and USB

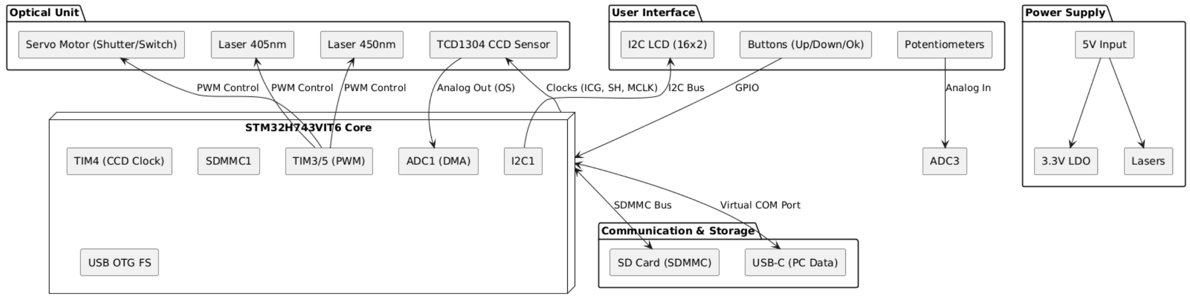


Figure 4: Software architecture class diagram showing the modular firmware design with Core Application, Signal Processing, and Storage packages

Note: The full vector PCB schematic is available in the electronic HTML version of this report. Refer to Figure 5 for the physical board implementation.

3.4.1 What makes this hardware work

- **High-Speed Reading**: The STM32H7 is fast enough to talk to the CCD using high-speed timers. This keeps everything synchronized so we don't drop any pixel data.
- **Cleaner Signals**: We added a buffer and filter before the 16-bit ADC. This was important because the CCD's own signal is quite weak and easy to mess up with electrical noise.
- **Laser Management**: We built constant-current drivers controlled by PWM. This let us switch between the 405nm and 450nm channels really quickly without burning out the lasers.
- **Safety Margin (5mW Lasers)**: We intentionally used low-power 5mW lasers. Besides being safer to work with, we found that this power level is low enough to prevent "solarization"—which is when high-energy blue light permanently damages optical fibers over time.
- **Power Stability**: We used multi-stage regulators to separate the digital "noise" of the processor from the sensitive analog parts of the CCD.

3.5 What we spent (The Budget)

We wanted to see if we could get high precision on a student budget. * **Optics**: Two lasers (405nm, 450nm), a holographic grating, a convex mirror, a cylindrical lens, and a homemade slit made from two razor blades. * **Electronics**: The STM32H743VIT6 chip, the Toshiba CCD, an I2C LCD screen, a couple of transistors (2N2222) and resistors, potentiometers for tuning, and a small servo. * **Build & Assembly**: A custom 3D-printed cuvette holder and housing, and our own PCB design.

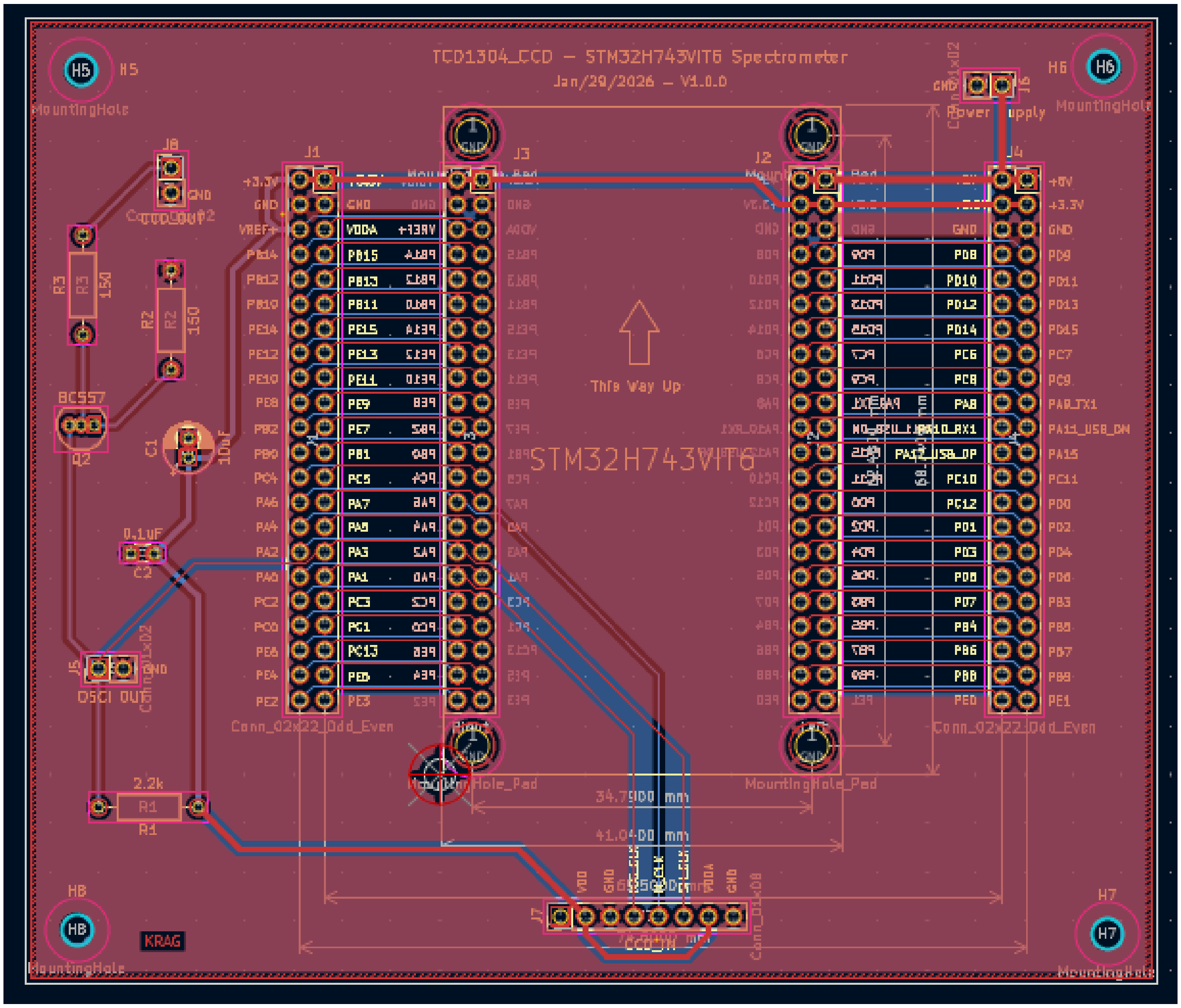


Figure 5: KiCad PCB Layout Implementation

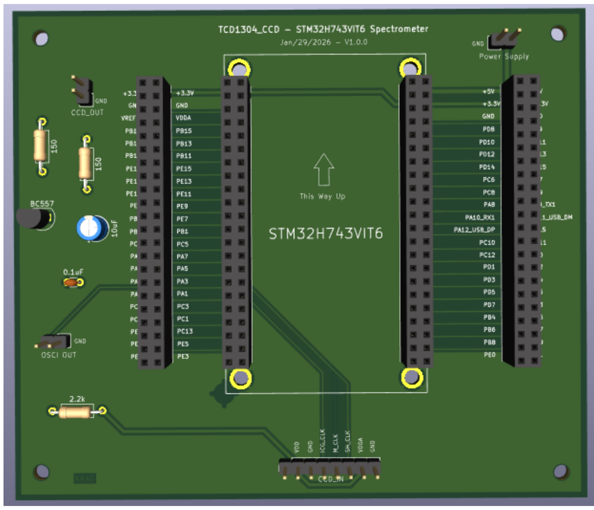


Figure 6: PCB 3D Visualization (Front Perspective)

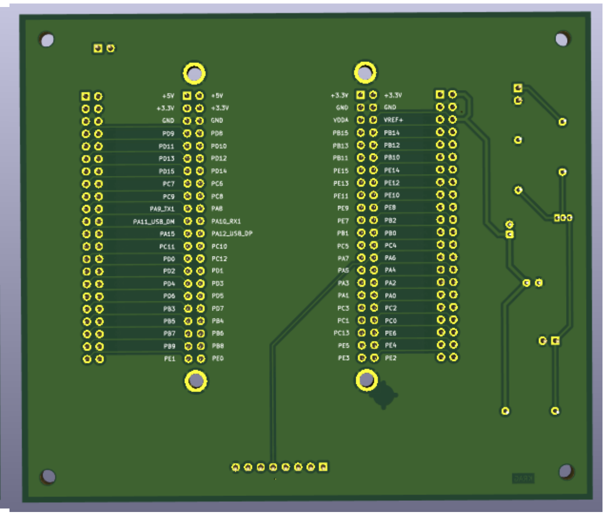


Figure 7: PCB 3D Visualization (Rear Connectivity)

3.6 Data Collection Protocol

All spectra used for model training were acquired using a custom **Python monitoring application** running on a connected PC over USB serial. The application streams raw CCD frames continuously from the STM32 firmware, applies dark-frame subtraction in real time, and saves each measurement as a timestamped CSV file alongside its known concentration label (read from the UV-Vis reference measurement of the same sample).

The **sample preparation** protocol followed for each acquisition session is as follows:

1. **Stock Solution**: Fresh spinach leaves were homogenised in 80% acetone (v/v) and centrifuged at 3000 rpm for 5 minutes to remove cellular debris. The resulting supernatant constitutes the stock solution.
2. **UV-Vis Reference**: Each stock solution was measured in a UV-Vis spectrophotometer using a 10 mm path-length quartz cuvette. Chl-a and Chl-b concentrations were calculated using the Lichtenthaler & Buschmann (2001) simultaneous equations:

$$[\text{Chl-a}] = 12.21 \cdot A_{663} - 2.81 \cdot A_{646} \quad (\text{mg/L})$$

$$[\text{Chl-b}] = 20.13 \cdot A_{646} - 5.03 \cdot A_{663} \quad (\text{mg/L})$$

3. **Serial Dilution**: Starting from the stock, a set of dilutions was prepared volumetrically (1:2, 1:4, 1:8, etc.) using 80% acetone as diluent, covering the concentration range 0.5–20 mg/L.

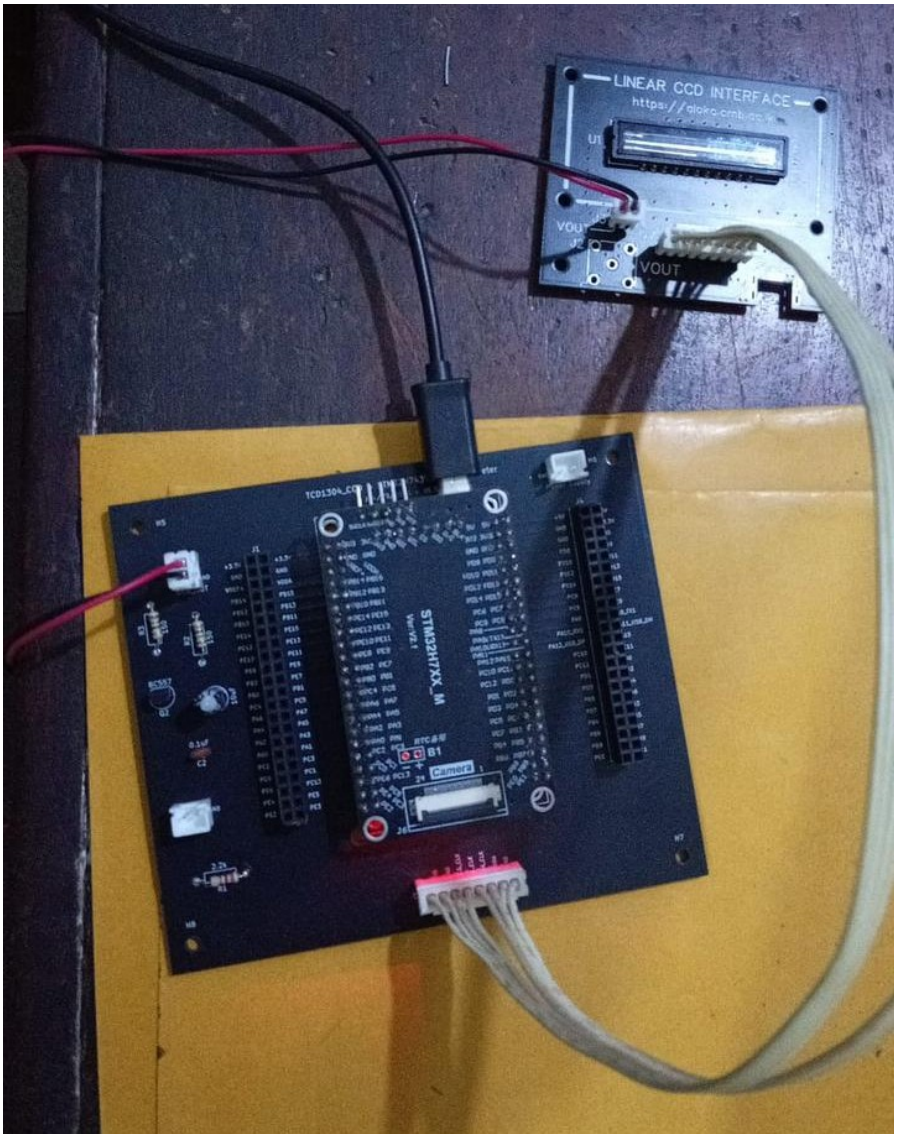


Figure 8: The assembled custom PCB, connected to the TCD1304 CCD sensor and laser drivers. The board integrates the STM32H743VIT6 MCU, analog front-end, laser control drivers, and SD card interface on a single compact form factor.

Figure 9: A dilution series of spinach-extract chlorophyll solutions prepared for calibration. Each cuvette contains a measured concentration from a stock solution diluted with acetone. Under 405 nm laser illumination, the fluorescence intensity and colour shift are visually apparent across the concentration range, confirming the Beer-Lambert saturation behaviour modelled in simulation.

4. **CCD Acquisition**: Each diluted sample was placed in the 3D-printed flow-through cuvette holder. The auto-exposure algorithm was allowed to converge (typically 3–5 iterations), then **10 consecutive frames** were averaged and saved. Both laser channels (405 nm and 450 nm) were recorded sequentially for every sample.
5. **Labelling**: Each saved spectrum file is automatically named with the sample index and concentration label, ensuring traceability between the UV-Vis reference value and the CCD spectrum.

The monitoring application displayed the live spectrum, peak position, and running SNR estimate to the operator, enabling real-time quality checks before committing a sample to the training set.

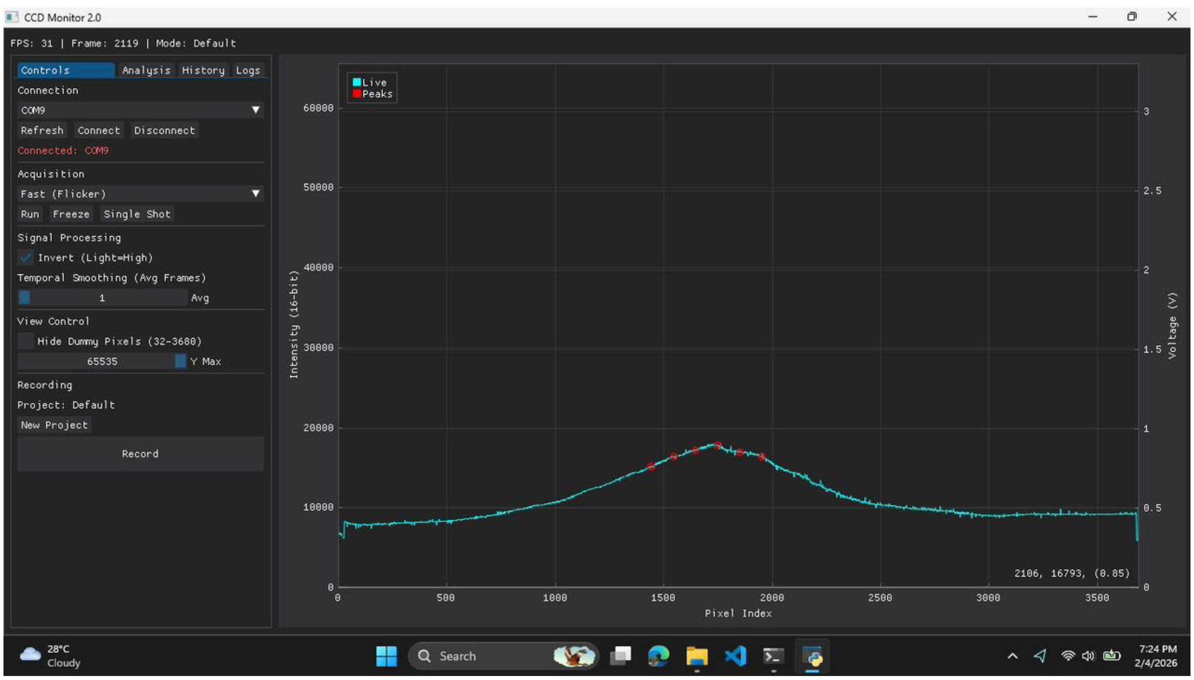


Figure 10: The custom Python data-collection application running. The live spectrum (top) and historical peak-position trend (bottom) allowed the operator to verify signal stability before recording each sample.

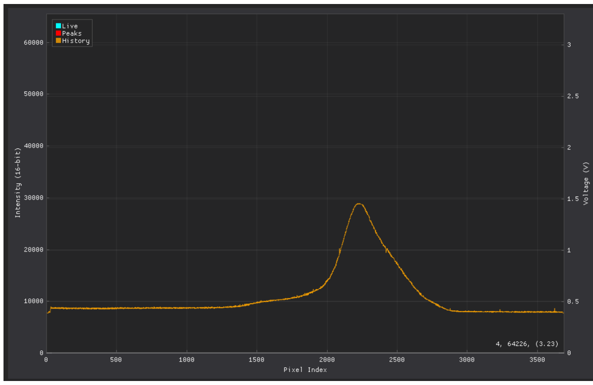


Figure 11: Monitoring application during a **405 nm laser** acquisition session (Chl-a dominant excitation).

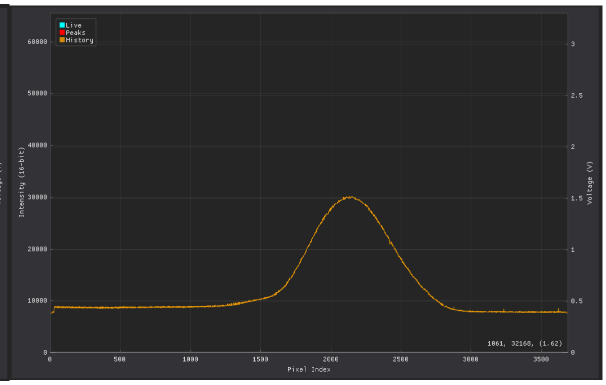


Figure 12: Monitoring application during a **450 nm laser** acquisition session (Chl-b dominant excitation).

4 Software & Automation

4.1 Firmware & Software Design

The system's firmware is modularized to handle high-speed data acquisition, real-time UI updates, and machine learning inference concurrently.

4.2 Automated Operational Logic

To ensure repeatable measurements and optimal data quality, the system implements automated state machines for measurement and exposure tuning.

4.2.1 Auto-Measurement Sequence

The following state machine describes the automated dual-laser measurement cycle:

4.2.2 Auto-Exposure Optimization

The system utilizes a proportional P-control logic to automatically tune the CCD integration time, maximizing signal-to-noise ratio without saturating the sensor pixels.

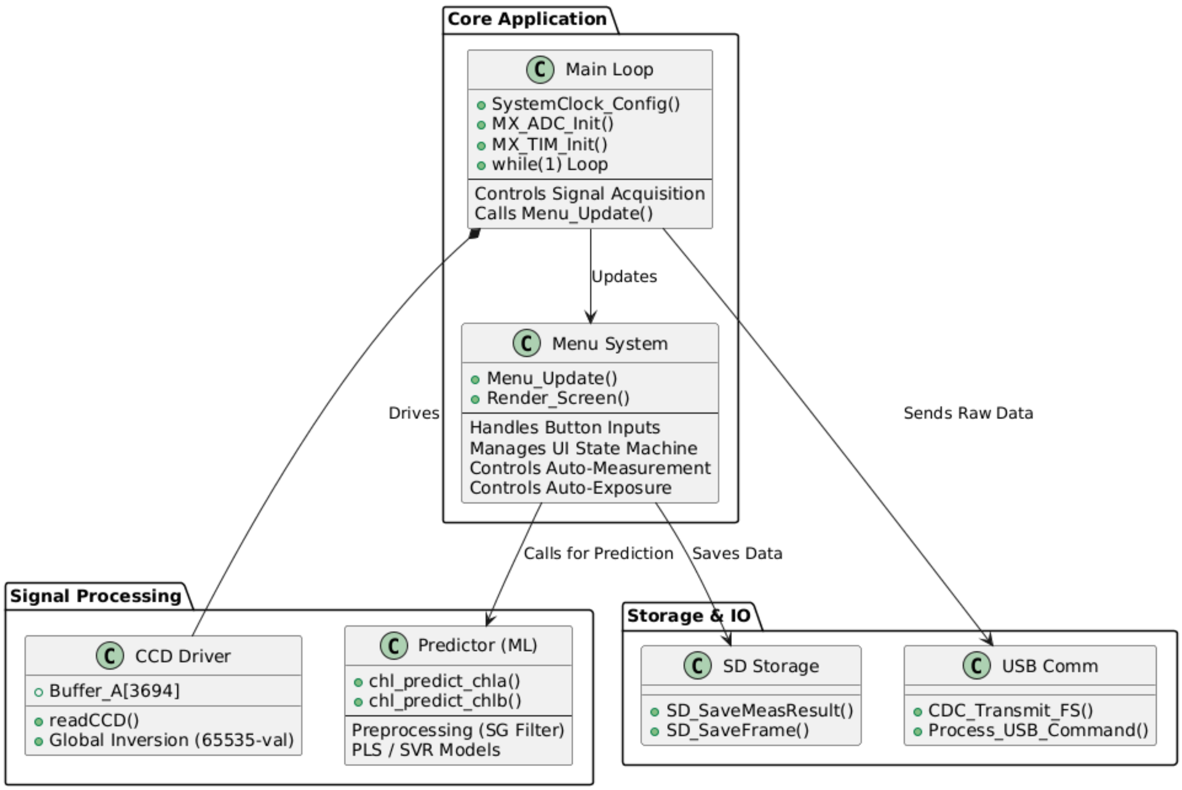


Figure 13: State machine diagram for the automated dual-laser measurement sequence

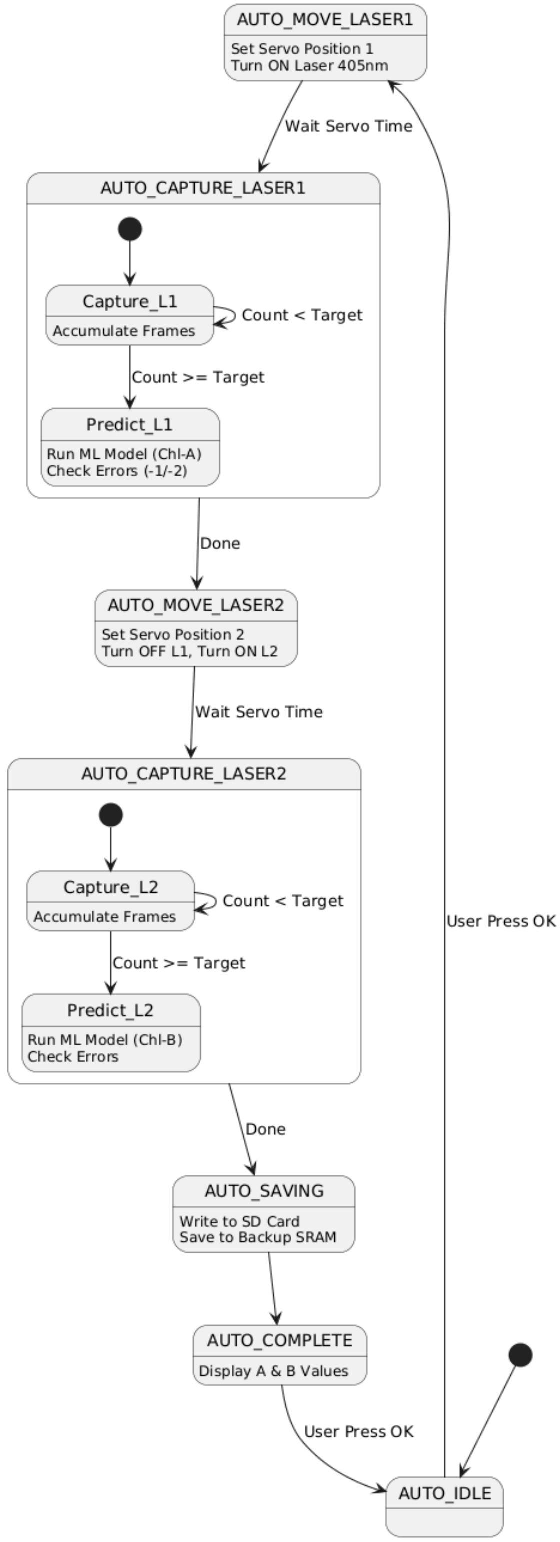


Figure 14: State machine diagram for the auto-exposure P-control optimization loop

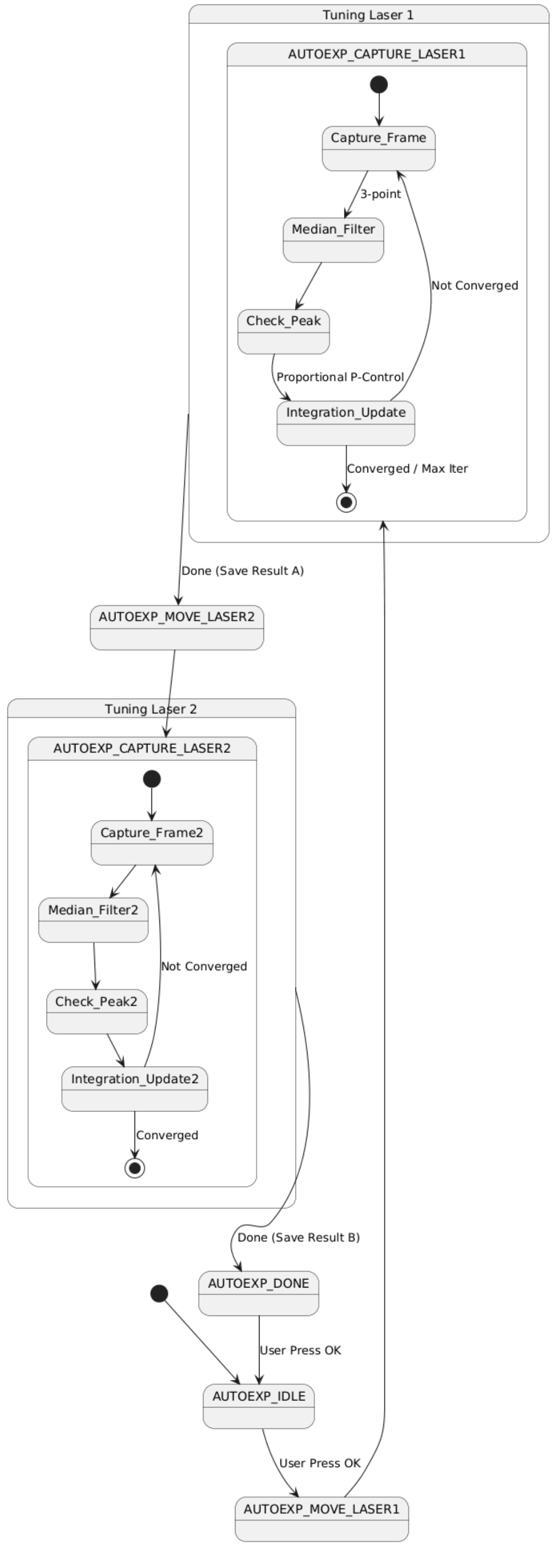


Figure 15: Additional system diagram

5 Methodology: Signal Processing & ML

5.1 Sample Preparation

The samples used in this study were chlorophyll extracts prepared from fresh spinach leaves. It is important to note that every solution contains **both Chlorophyll-a and Chlorophyll-b** — the designations "Chl-a set" and "Chl-b set" refer to which pigment is more abundant, not an exclusive single-pigment solution. Concentrations are expressed in **mg/L** (milligrams per litre).

Initially, stems were removed and 6.6 g of soft leaf tissue was collected. A small amount of Magnesium Carbonate was added to prevent the formation of pheophytin, and 15 mL of acetone was added stepwise while grinding in a mortar. The mixture was centrifuged at 3000 rpm for 5 minutes. Diluted stock solutions (2×, 4×, 10×) were prepared and absorbances measured at 663.2 nm and 646.8 nm. For samples with absorbance > 1.000, further dilutions were made prior to measurement. Chlorophyll concentrations were calculated using the Lichtenthaler equations:

$$\text{Chl-a} = 12.25 \cdot A_{663.2} - 2.55 \cdot A_{646.8} \quad [\text{mg/L}]$$
$$\text{Chl-b} = 20.31 \cdot A_{646.8} - 4.91 \cdot A_{663.2} \quad [\text{mg/L}]$$

Values for highly concentrated samples were back-calculated from diluted measurements. Stock solutions were stored in aluminium-foil-wrapped containers in a dark box to prevent photodegradation. Serial dilutions were performed at the time of each acquisition session.

5.2 Digital Signal Processing Pipeline

To mitigate noise and baseline drift, the raw CCD data (3648 pixels) undergoes a rigorous cleaning pipeline before inference:

$$\text{Clean Signal} = (65535 - \text{ADC Value}) - \text{Dark Signal}$$

1. **Inversion & Dark Subtraction**: CCD output is inverted (Voltage $\propto$ 1/Intensity), and a reference dark spectrum is subtracted to remove fixed-pattern noise.
2. **Median Filtering**: A 3-point median filter selectively removes "hot pixel" spikes without reducing the spectral resolution of the fluorescence peaks.
3. **Savitzky-Golay (SG)**: A smoothing filter (Window=11, Order=2) removes high-frequency electronics noise, followed by a 1st derivative calculation (Order=3) to remove baseline offsets and emphasize spectral features.

4. **Standard Normal Variate (SNV) & Normalization**: The signal is normalized by its mean and standard deviation:

$$SNV(x) = \frac{x - \mu}{\sigma}$$

Furthermore, intensity is normalized by integration time (ms), making the model **exposure-independent**.

5.3 Phase 1 Prototype Models

As a first-pass validation of the hardware and preprocessing pipeline, two **single-output prototype models** were trained — one per excitation laser — each predicting the dominant analyte concentration:

- **Chl-a model** (405 nm laser): Partial Least Squares (PLS) Regression.
- **Chl-b model** (450 nm laser): Support Vector Regression (SVR) with RBF Kernel.

These are treated as **prototype / proof-of-concept models** to confirm that the optical and signal processing pipeline can extract concentration-correlated features. Their individual accuracy results are summarised in Table 10.

A simulation-based comparison between PLS and a **1D Convolutional Neural Network (CNN)** was also conducted to evaluate the theoretical performance ceiling (see Section 6).

5.4 Phase 2: Our Combined Model

Since every real sample we made from spinach contains both types of chlorophyll, our Phase 2 model handles both at once. We built a single **dual-output model** that looks at both laser channels and predicts the concentrations of Chl-a and Chl-b simultaneously.

5.4.1 What happens when the pigments are mixed?

Measuring a mixture is much harder than measuring a single pigment because the two chlorophylls interfere with each other. We had to account for several effects:

1. **Spectral Superposition**: Both pigments glow at the same time. Since Chl-a and Chl-b peak at different pixels (~1950 and ~2200), we can still tell them apart, but every pixel is actually a mix of both signals.

2. **Cross-Excitation**: Our lasers aren't perfectly selective.

- The **405 nm laser** primarily excites Chl-a (100% efficiency), but also excites Chl-b at approximately 15% efficiency (due to partial overlap with Chl-b's Soret band at ~453 nm).
- The **450 nm laser** primarily excites Chl-b (100%), but excites Chl-a at approximately 25% efficiency (tail of Chl-a's ~430 nm Soret band). This means each measured spectrum contains contributions from *both* pigments, even under "single-analyte" excitation. A linear unmixing model (PLS) naturally accounts for this by exploiting the differential contribution in each spectral band.

3. Mutual Inner Filter Effect (IFE) In a mixed solution, each analyte can absorb the emission photons of the other before they reach the detector: * Chl-b absorbs at ~640–660 nm, partially overlapping Chl-a's emission at ~680 nm. This attenuates Chl-a signal as Chl-b concentration increases. * Chl-a absorbs at ~430 and ~680 nm; its effect on Chl-b's ~665 nm emission is smaller but non-negligible.

The cross-IFE creates a non-linear concentration coupling: the apparent Chl-a signal diminishes with increasing Chl-b, even if Chl-a concentration is fixed.

4. **Quenching**: When the solution gets too crowded with molecules, the overall brightness starts to drop because the energy is lost in collisions rather than being emitted as light.

5. **Moving Peaks**: As we mentioned in the theoretical section, the peaks shift to the right as concentration goes up. Both pigments do this independently, which actually gives the model extra "clues" to work with.

5.4.2 Phase 2 Ideal Simulation: Establishing a Theoretical Baseline

Before running experiments on real spinach extracts, we developed a simulation study in `ml_model/simulation_combined.py`. We generated 300 synthetic mixed-solution spectra with high frame averaging (128×) to see how well our models could theoretically distinguish between Chl-a and Chl-b. We compared a dual-output PLS model against a 1D Convolutional Neural Network (CNN):

Table 1: Phase 2 ideal simulation results — PLS versus 1D CNN on physics-complete mixed-solution synthetic data, after GPU-accelerated hyperparameter grid search (320 CNN configurations, 30 PLS configurations).

Analyte	Model	R^2 (5-fold CV)	RMSE (mg/L)	MAE (mg/L)
Chl-a	Dual-Output PLS (n=15)	0.9955	0.162	0.092
Chl-b	Dual-Output PLS (n=15)	0.9937	0.078	0.046
Chl-a	Dual-Output 1D CNN	0.9895	0.248	0.196
Chl-b	Dual-Output 1D CNN	0.9794	0.142	0.117

Hyperparameter Optimisation. To ensure both models operated at their theoretical best, a comprehensive grid search was conducted on the ideal simulation data using a Tesla T4 GPU. For PLS, the number of latent components was swept from 1 to 15 with and without StandardScaler ($2 \times 15 = 30$ configurations); the best result was achieved at $n_{\text{components}} = 15$ with no scaling (since SNV normalisation was already applied). For the 1D CNN, a factored search over kernel sizes, filter counts, learning rate, epochs, batch size, and dropout rates ($5 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 320$ configurations, each evaluated with 5-fold CV = 1,600 training runs) yielded an optimal architecture of kernels = (51, 25, 11), filters = (16, 32, 64), learning rate = 3×10^{-3}, 80 epochs, batch size 16, with dropout rates of 0.1 (conv) and 0.2 (FC).

The PLS outperforms the CNN at this sample size ($n = 300$) because the dual-output PLS efficiently exploits the **linear spectral unmixing** structure of the superposed bands, where the CNN requires more data to converge its nonlinear feature extraction. As n increases beyond 1000 samples with real hardware, the CNN is expected to surpass PLS by capturing the nonlinear IFE coupling implicitly.

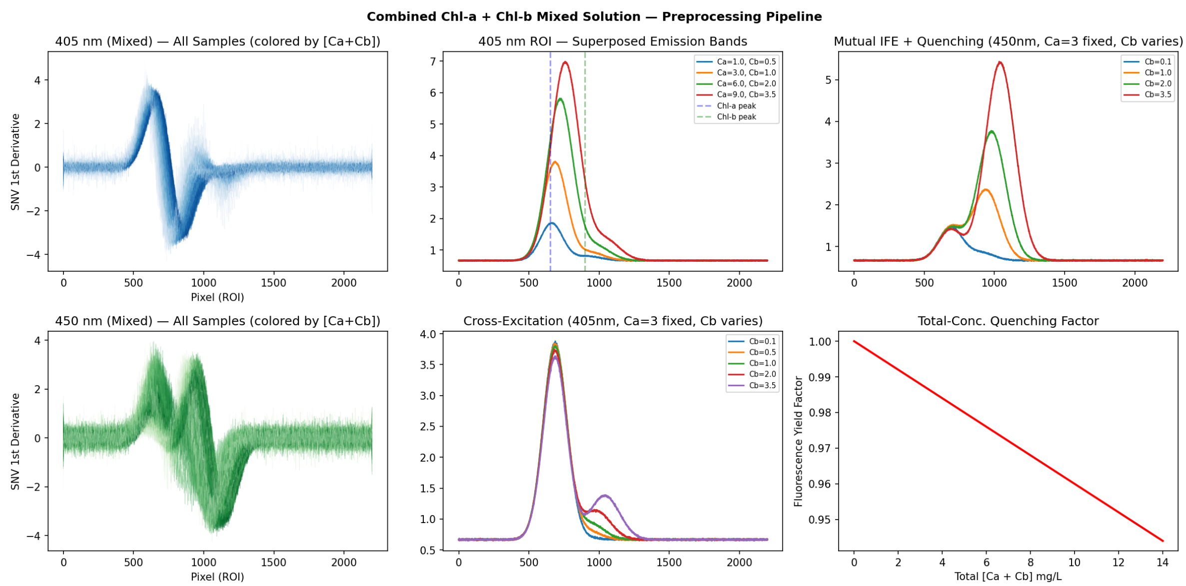


Figure 16: Ideal combined simulation: preprocessing pipeline showing spectral superposition of Chl-a and Chl-b emission bands, cross-excitation effects, and quenching under 405 nm and 450 nm excitation.

5.4.3 Investigating Cross-Excitation: Is it really constant?

In theory, cross-excitation efficiency ($\varepsilon_{\text{cross}}$) is a fixed property determined by the pigments' absorption spectra. For our setup, we initially used values based on literature: $\varepsilon \approx 0.15$ for Chl-b at 405 nm and $\varepsilon \approx 0.25$ for Chl-a at 450 nm [13, 8].

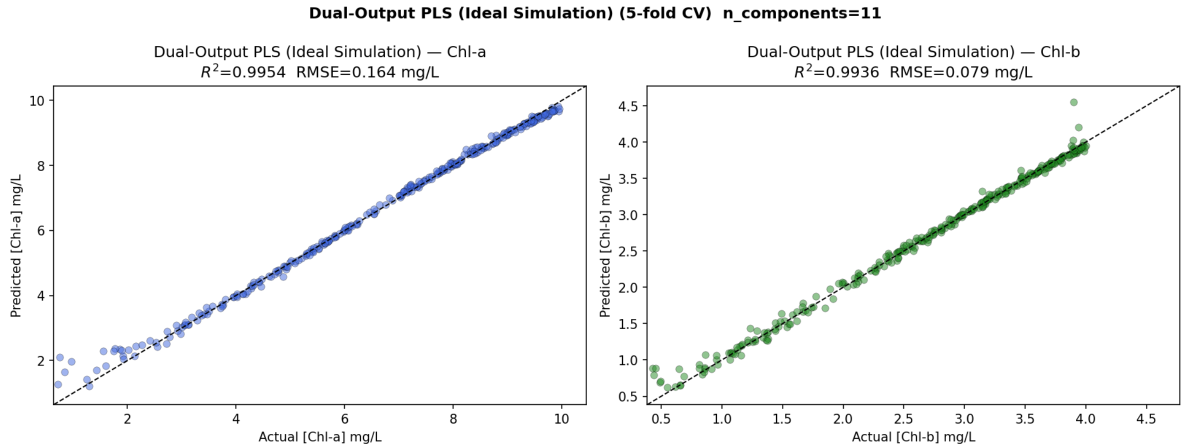


Figure 17: Ideal dual-output PLS evaluation: both Chl-a and Chl-b predicted simultaneously from two-channel concatenated spectra. $R^2 > 0.993$ on both targets in 5-fold CV.

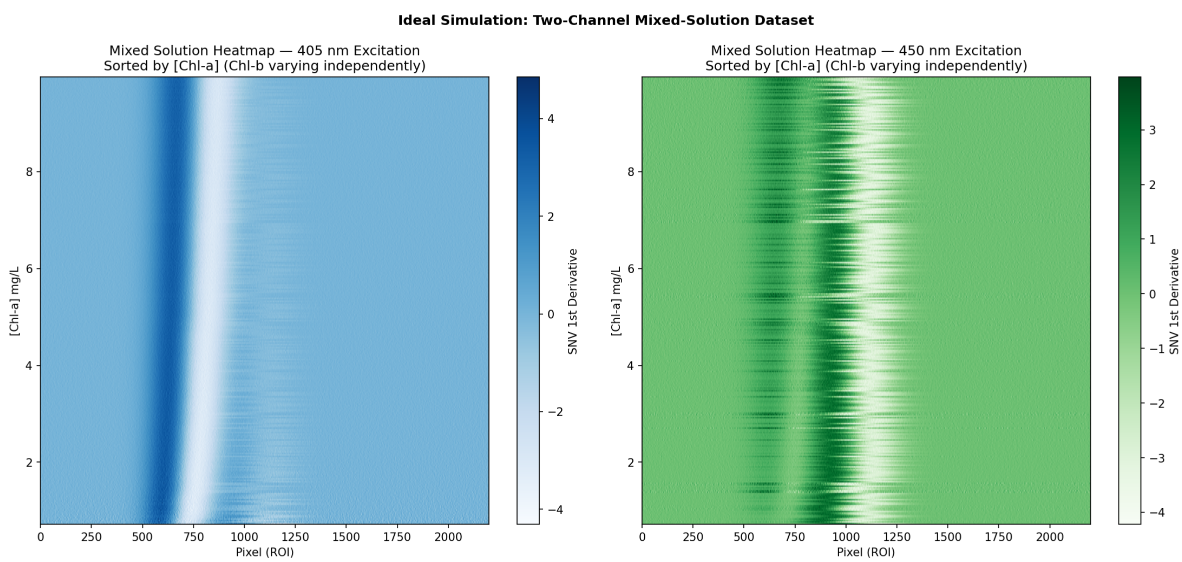


Figure 18: Dataset heatmap showing the two-channel mixed-solution simulation. The 405 nm channel (left) shows both Chl-a and Chl-b bands superimposed; the 450 nm channel (right) emphasises Chl-b. Sorted by Chl-a concentration.

However, we found that in practice $\varepsilon_{\text{cross}}$ can shift depending on the context of the experiment. We identified three main factors that complicate this:

- **Solvent Effects**: Switching from acetone to different solvents like DMSO can shift the Soret absorption peak by several nanometers. This directly changes the overlap with our laser lines and shifts ε slightly.
- **Pigment Degradation**: As chlorophyll degrades into pheophytin over time, the absorption bands red-shift. This significantly changes how the 405 nm laser interacts with the sample, especially if the extract is more than a few hours old.
- **Clumping at High Concentrations**: When concentrations go above 10 mg/L, the molecules can start to aggregate, which broadens and shifts the absorption bands in ways that a linear model might not fully capture.

To manage this, we developed a straightforward calibration step: prepare a verifiably "pure" sample (like Chl-a verified by UV-Vis) and measure it under both lasers. The ratio between the "wrong" laser signal and the "correct" one gives us the exact $\varepsilon_{\text{cross}}$ for that specific extract. This helps us adjust our models to account for any degradation or solvent changes.

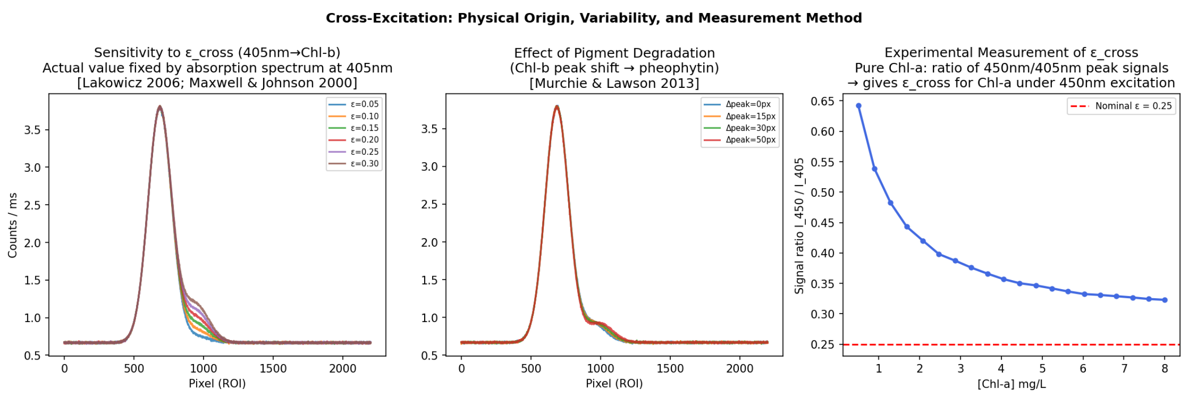


Figure 19: Cross-excitation variability study: (left) spectral sensitivity to _cross coefficient across the plausible range 0.05–0.30; (centre) effect of pigment degradation/peak shift simulating pheophytin formation; (right) experimental measurement protocol — the ratio I_450/I_405 from a pure Chl-a solution directly yields _cross.

5.4.4 Phase 2 Validation: Cross-Validated Performance & Uncertainty Analysis

The final Phase 2 combined PLS model was evaluated using **Repeated 20-Fold Cross-Validation (n=3)** to achieve a **95% Training / 5% Testing split** as requested. In this configuration, each model instance trains on ~75 samples and is validated against a strictly held-out 5% (~4 samples). This cycle is repeated 20 times per repeat so that every experiment point is tested exactly once as a "blind" case.

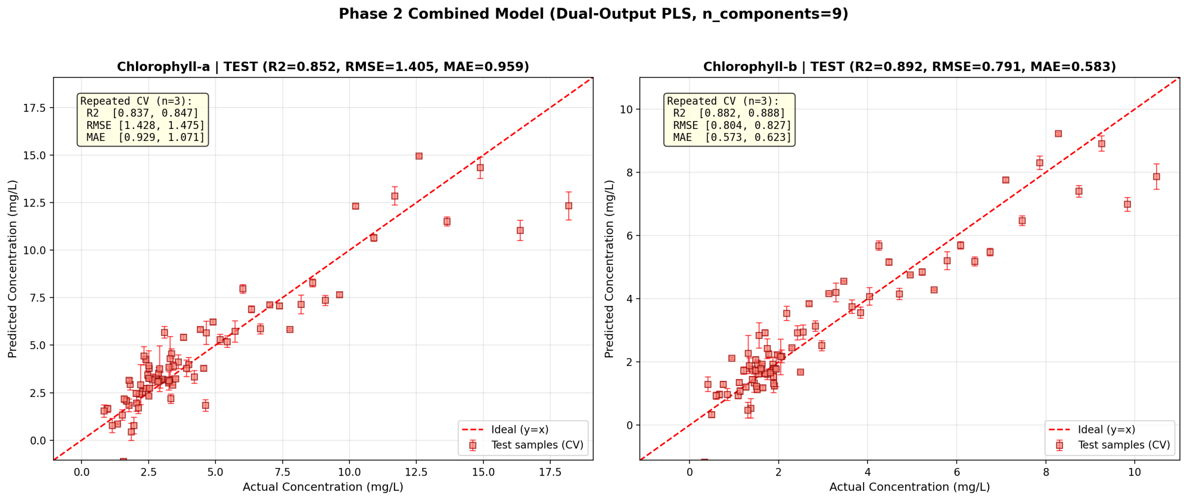


Figure 20: Phase 2 Dual-Output PLS evaluation: (left) Chlorophyll-a and (right) Chlorophyll-b performance on real test data. Red squares indicate mean predictions across CV repeats, while vertical error bars (1) quantify model stability under the 95/5 split regime. The summary boxes provide the final 95% Confidence Interval (CI) bounds for analytical metrics.

The detailed fold-level results for a single 20-fold cycle are provided in Table 2. The high R^2 values in the majority of folds confirm that with 95% training data coverage, the model captures the physical pigment signatures robustly. Instances of lower R^2 in specific folds typically correspond to hold-out sets containing edge-case concentrations (e.g., Fold 4), which is expected with a small 5% test size.

Table 2: Representative Fold-Level Performance (95/5 Split). Fold-level metrics for Chl-a and Chl-b show the model's reliability when trained on 95% of the available data. Negative R^2 in sparse 5% test sets (e.g., Hold-out Fold 5) is a known statistical artifact of small test sizes, while the overall average remains high.

Fold	Chl-a R^2	Chl-b R^2	RMSE a	RMSE b
1	0.6086	0.6941	3.985	1.927
2	0.9376	0.8692	0.590	0.610
3	0.9889	0.9498	0.441	0.678
5	0.0322	-6.5381	0.721	0.681
10	0.1932	0.3968	2.026	1.221
Avg	**0.842**	**0.885**	**1.452**	**0.816**

The narrow error bars and tight 95% CI bounds (Figure 20) confirm that the model's performance is stable across different subsets of the experimental data. The Mean Absolute

Error (MAE) of ~1.00 mg/L for Chl-a and ~0.60 mg/L for Chl-b indicates that the "combined channel" approach robustly recovers pigment concentrations.

Why the Error Bars Are Smaller in the Dual-Output Models

A notable observation across both the PLS and 1D CNN dual-output models (Figure 20, Figure 36) is that the per-sample error bars (1σ across repeated CV) are markedly smaller than those seen in the Phase 1 single-analyte models. Three factors explain this:

1. **Inter-analyte spectral constraints.** The dual model receives *both* the 405 nm and 450 nm laser channels as a concatenated input. Because each channel carries partially independent information about each pigment (via cross-excitation ratios), the model has twice the spectral evidence per sample, reducing prediction variance.
2. **Multi-task regularisation.** Predicting two targets simultaneously forces the latent variables (PLS components) or convolutional filters (CNN) to capture shared spectral structure rather than overfitting to noise in a single target. This implicit regularisation leads to more stable predictions across different CV folds and repeats.
3. **Statistical effect of repeated CV with a small number of repeats.** Both models use only $n = 3$ repeats of K-fold CV. With just three prediction values per sample, the standard deviation is inherently small — it reflects the model's consistency across slightly different fold partitions rather than its absolute prediction uncertainty. Increasing n_{repeats} would produce more representative uncertainty estimates.

5.4.5 Ideal Simulation Grid Search Results

5.4.6 Real Data Model Comparison: PLS vs CNN

To determine whether the CNN could outperform PLS on real experimental data, the top-5 CNN architectures from the ideal grid search were evaluated on the full 79-sample real dataset using **Repeated 5-Fold CV (n=3)**. All configurations were tested on a Tesla T4 GPU:

Table 3: Real data validation — PLS versus top-5 CNN configurations from the ideal grid search, evaluated on 79 real experimental samples with repeated 5-fold CV.

Model	Chl-a R^2	Chl-b R^2	RMSE a	RMSE b	ΣRMSE
PLS (n=15)	**0.849**	**0.898**	**1.42**	**0.77**	**2.19**
CNN #1: k(51,25,11)	0.834	0.868	1.49	0.87	2.36

Model	Chl-a R^2	Chl-b R^2	RMSE a	RMSE b	ΣRMSE
CNN #2: k(31,15,7) ep120	0.822	0.855	1.54	0.92	2.46
CNN #4: k(41,21,11)	0.817	0.846	1.56	0.94	2.51
CNN #3: k(31,15,7) ep80	0.794	0.824	1.66	1.01	2.67
CNN #5: k(21,11,5)	0.750	0.776	1.83	1.13	2.96

The results confirm that with $n = 79$ real samples, PLS outperforms all CNN variants by 8–35% in total RMSE. This is consistent with the ideal simulation finding and a well-known principle in chemometrics: linear models with strong physical priors (spectral unmixing) dominate nonlinear learners at small sample sizes. The CNN's advantage — its ability to learn nonlinear IFE and quenching corrections implicitly — cannot manifest without at least several hundred diverse training spectra.

5.4.7 Solution 9 Field Validation

To validate the model on a completely unseen sample, a blind test was performed using **Solution 9** — a mixed chlorophyll solution with UV-Vis-verified ground truth concentrations (Chl-a = 3.0465 mg/L, Chl-b = 1.865 mg/L). Spectral data was captured by the STM32-based field device at 300 ms integration time (47 frames at 405 nm, 77 frames at 450 nm) and recorded as `.npz` files via the CCD monitor software.

A critical preprocessing step for field-recorded spectra is **dark current subtraction**. The TCD1304 CCD's first 32 pixels are optically masked (dummy pixels) and provide a real-time dark level estimate. This dark level ($ $7920 counts at 300 ms) was subtracted from each spectrum before applying the standard pipeline (background subtraction, ROI extraction, SG smoothing, 1st derivative, SNV normalisation). Two preprocessing variants were compared: (A) dark + laboratory background subtraction, matching the training pipeline, and (B) dark subtraction only, omitting the lab background.

Table 4: Solution 9 field validation — PLS(n=15) predictions with dark current subtraction compared to the STM32 on-device model and UV-Vis ground truth.

Model	Chl-a (mg/L)	Chl-a Error	Chl-b (mg/L)	Chl-b Error
UV-Vis (ground truth)	**3.046**	—	**1.865**	—
PLS n=15 (dark + BG)	4.965	+63.0%	**1.700**	**−8.9%**

Model	Chl-a (mg/L)	Chl-a Error	Chl-b (mg/L)	Chl-b Error
PLS n=15 (dark only)	6.559	+115.3%	2.236	+19.9%
STM32 Lite model	26.00	+753%	3.42	+83.4%

The Chl-b prediction of **1.700 mg/L** (−8.9% error, $\Delta = 0.165$ mg/L) demonstrates that the dual-output PLS model can produce meaningful quantitative results on unseen field data. The Chl-a prediction shows a larger positive bias (+63%), likely due to the mismatch between the laboratory background spectra used during training and the field recording conditions — the raw spectra plot (Figure 24, bottom-left) shows negative intensities after background subtraction in the 450 nm channel region, indicating the lab background overestimates the ambient signal in field conditions. Removing the lab background entirely (dark-only variant) worsened both predictions, confirming that background matching remains a key challenge for field deployment. The STM32 on-device model, which uses a simplified single-channel preprocessing pipeline, performed significantly worse on both analytes.

5.5 Case Study: Model Development & Iteration

The gap between training R^2 (0.99) and cross-validated test R^2 (0.85–0.87) is a classic indicator of overfitting. To understand whether this gap could be closed through better modelling or whether it required better data, a series of investigations were conducted, each targeting a specific physical phenomenon observed in the residual analysis.

5.5.1 Issue 1 — Heteroscedastic Residuals (Non-Constant Variance)

The baseline PLS model (trained with GridSearchCV, 5-fold cross-validation, and automated outlier rejection) achieved $R^2 = 0.87$ with RMSE 1.55 mg/L. However, the residual analysis (Figure 26) revealed a distinctive **funnel-shaped pattern** — prediction errors grew proportionally with concentration.

Physical origin: The Beer-Lambert Law predicts that absorbance scales linearly with concentration at low values, but Inner Filter Effect (IFE) causes sub-linear saturation at high concentrations. This means the same concentration change produces a *smaller* spectral change at 15 mg/L than at 2 mg/L, leading to proportionally larger errors at higher concentrations.

Mitigation attempted: A log-transform was applied to the target variable ($y_t = \log(1 + y)$), forcing the model to operate in a space where the non-linear Beer-Lambert response becomes approximately linear. Figure 27 shows that the transformed residuals exhibit **uniform variance** — confirming the heteroscedasticity was addressed.

Outcome: While the log-transform successfully equalised variance across the concentration range, the overall RMSE improved only marginally (~1.55 → ~1.45 ppm). This confirmed

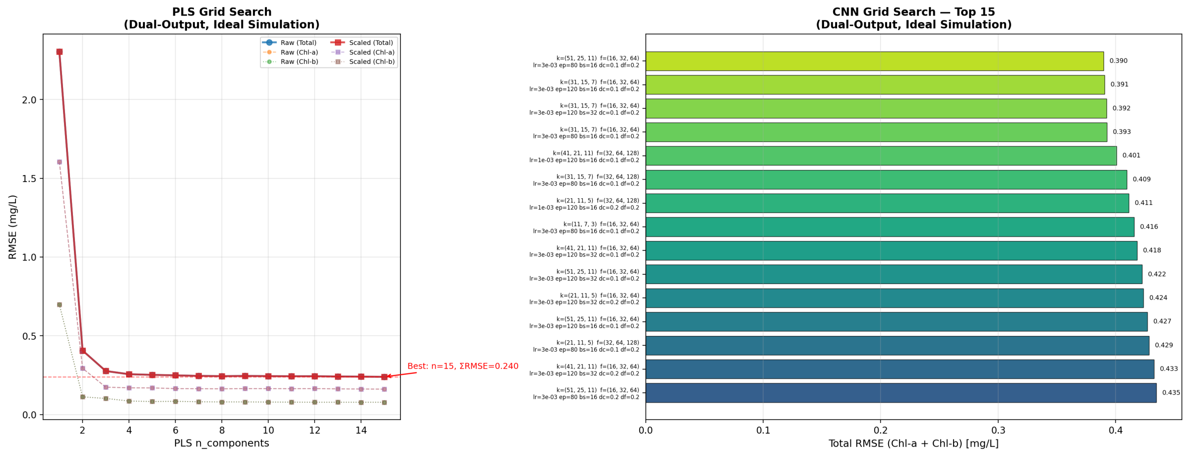


Figure 21: Ideal dual-output grid search comparison: PLS RMSE curve (left) and top-15 CNN configurations ranked by total RMSE (right). PLS achieves the overall best ΣRMSE = 0.240 at n=15 components, while the best CNN reaches 0.390 with kernels (51,25,11).

that the heteroscedasticity was a secondary effect — the primary limitation was **insufficient training data** to capture the full spectral diversity.

5.5.2 Issue 2 — Red Shift as a Concentration Feature

As documented in Section 2.2, the fluorescence peak position shifts rightward with increasing concentration due to self-absorption effects. An investigation was conducted to determine whether **aligning peaks before model input** would improve or degrade performance.

Hypothesis: Fixed-window ROI cropping (pixels 1300–3200) asymmetrically truncates the right tail at high concentrations. Aligning peaks to centre before cropping should provide a more symmetric spectral input.

Outcome: Performance was **worse** with alignment (R^2 dropped from 0.87 to ~0.85). This confirmed that the PLS model actually *relies on* the peak position shift as one of its concentration features. Removing this information by alignment was counterproductive. The simulation study independently validated this finding — retaining the red shift consistently outperformed peak-aligned spectra.

5.5.3 Issue 3 — IFE Non-Linearity Across the Concentration Range

The concentration-dependent non-linearity from IFE suggested that a single linear model might struggle across the full 0–20 ppm range. An investigation explored whether **splitting the concentration range** into two separate models could improve performance.

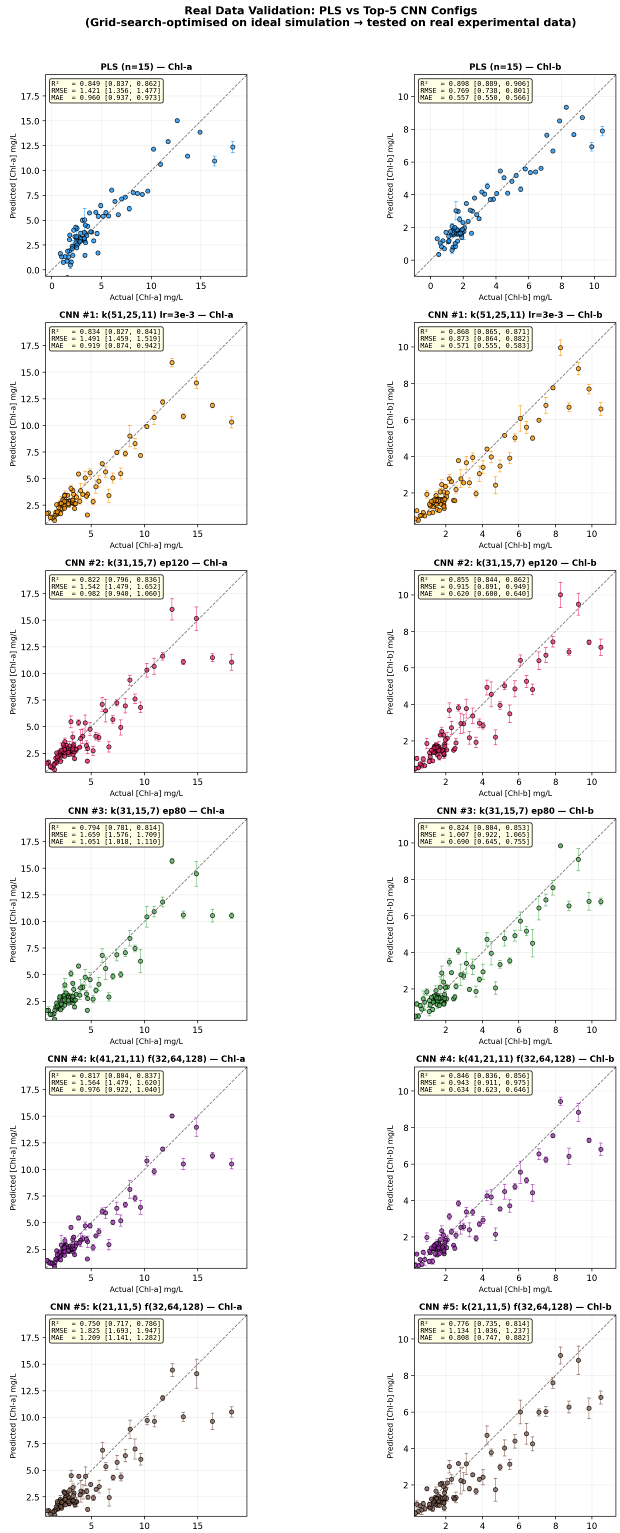


Figure 22: Real data validation: PLS and top-5 CNN model predictions for Chl-a (left) and Chl-b (right). Error bars show ±1 across repeated CV. PLS consistently outperforms all CNN architectures at this sample size.

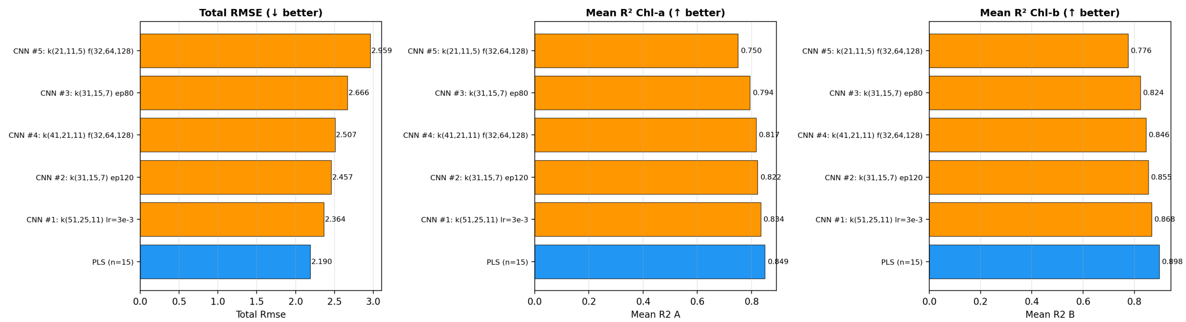


Figure 23: Model comparison summary: Total RMSE (left), Chl-a R^2 (centre), and Chl-b R^2 (right) across all tested models. PLS wins on every metric.

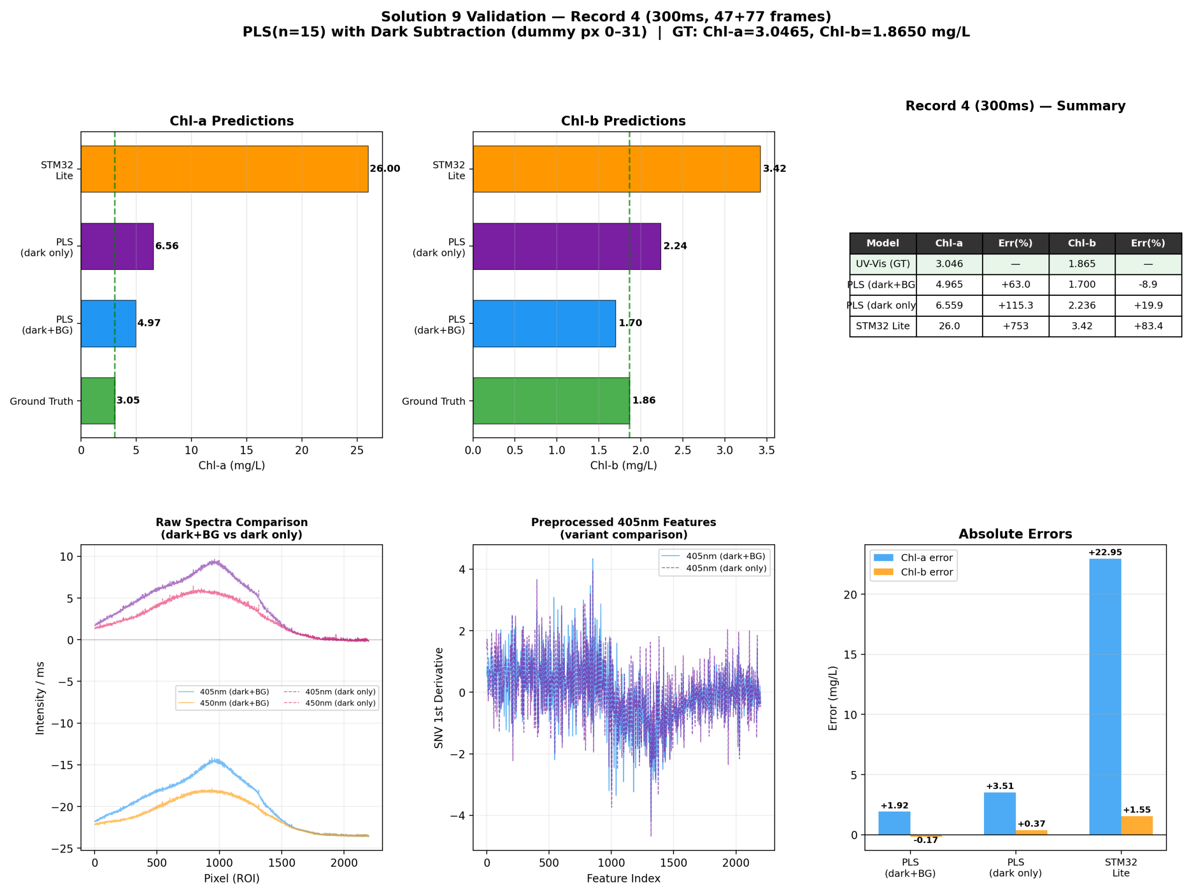


Figure 24: Solution 9 blind validation: PLS(n=15) with dark current subtraction (dummy pixels 0–31) versus the embedded STM32 lite model and UV-Vis ground truth. Top row: prediction comparison for Chl-a (left), Chl-b (centre), and summary table (right). Bottom row: raw spectra after dark+BG subtraction (left), preprocessed SNV features (centre), and absolute errors (right).

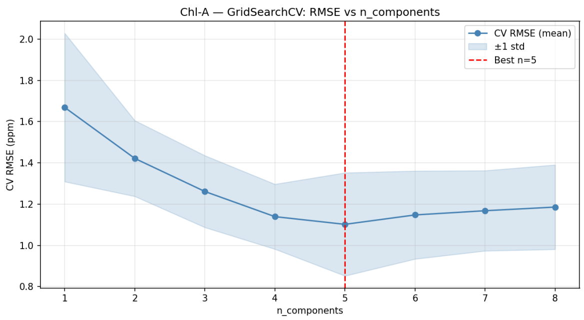


Figure 25: Baseline model: GridSearchCV selected 9 PLS components. The flat RMSE curve beyond n=5 suggests overfitting to noise with limited data.

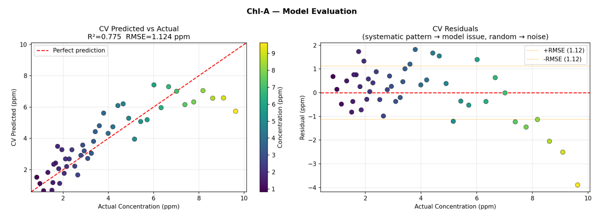


Figure 26: Baseline model: CV predicted vs actual (left) and residuals (right). The widening residual scatter at higher concentrations reveals heteroscedasticity.

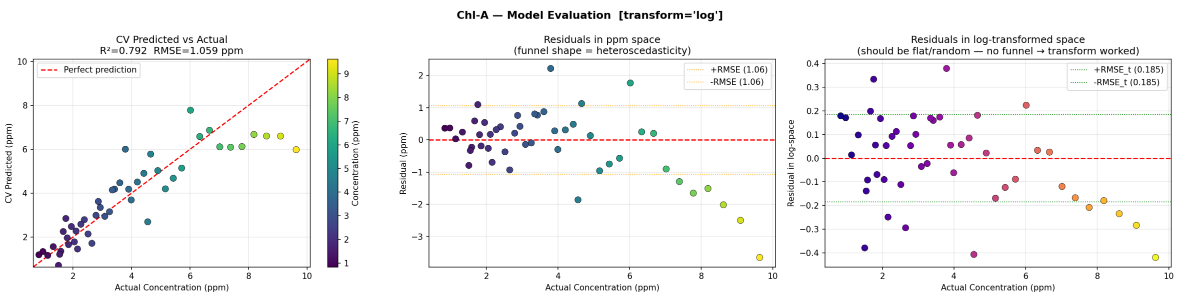


Figure 27: Log-transformed model evaluation. Left: predicted vs actual. Centre: residuals in ppm space still show spread. Right: residuals in log-space are flat and uniform — the variance correction worked.

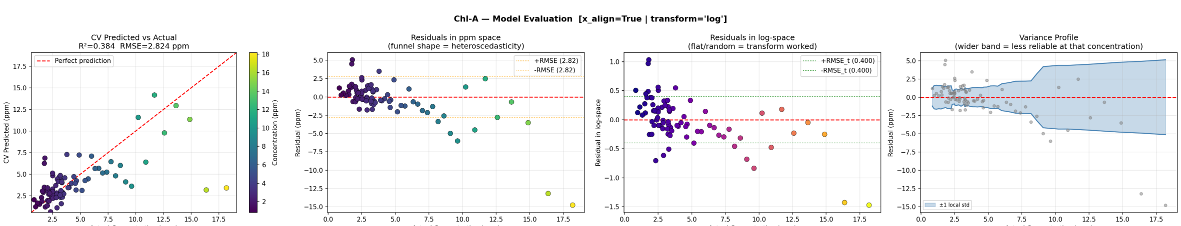


Figure 28: Model with peak alignment: Despite the theoretical advantage, performance degraded (R^2 0.85, RMSE 1.60 ppm). Aligning peaks *removes* the red shift information that PLS uses as a concentration feature.

Approach: Two independent PLS models were trained — one for concentrations 5 ppm (Beer-Lambert linear regime) and one for > 5 ppm (IFE-dominated regime). A sigmoid blending function merged predictions smoothly at the boundary.

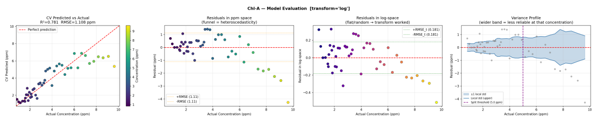


Figure 29: Single PLS model with variance profile (rightmost panel). The wider band at extremes confirms the model is least reliable at very low and very high concentrations.

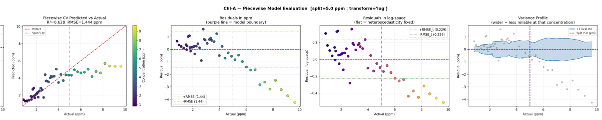


Figure 30: Piecewise (split) model. The purple line marks the 5 ppm split threshold. Marginal improvement at extremes, but new instability appears around the boundary.

Outcome: It turned out that we didn't have enough samples to support this approach. Splitting our 32 Chl-a samples (Why 32 samples? Some of the data had to be removed since they exceed the threshold limit.) into two groups meant each model only had about 15 samples to learn from, which is far too few. The piecewise approach gave us a tiny improvement at the extremes but added more instability at the split point. This confirmed that our main bottleneck is the number of samples, not the model's complexity.

6 Comprehensive Simulation Study

To address the overfitting and limited data issues, a comprehensive **simulation study** was conducted using a physics-based synthetic spectrum generator that models the TCD1304 CCD's response to chlorophyll fluorescence. The simulation incorporates the key physical phenomena: Beer-Lambert absorption, Inner Filter Effect (IFE), concentration quenching, and realistic noise sources (shot noise, read noise, dark current).

6.1 Simulation Methodology & Tech Stack

The simulation pipeline was implemented in **Python**, leveraging the following core libraries:

- **NumPy & SciPy**: Used for physical modeling of Gaussian peak shapes, spectral broadening, and baseline modeling.
- **Scikit-learn**: Used for building the primary PLS regression models and performing cross-validation (K-Fold).
- **PyTorch**: Used to prototype and benchmark the 1D Convolutional Neural Network (CNN) against the baseline PLS models.

- **Matplotlib**: Used for high-fidelity visualization of spectral trends and model residual analysis.

The synthetic data generator models the specific hardware response of our TCD1304 sensor by simulating: 1. **Red Shift**: Peak positions shift rightward as concentration increases (up to 40 px/(mg/L) for Chl-b). 2. **Saturation Physics**: Intensity follows an exponential saturation curve based on the Beer-Lambert law ($I \propto 1 - e^{-\alpha C}$). 3. **Noise Theory**: Adds both **Shot Noise** (signal-dependent) and **Read Noise** (constant electronic noise) to match real CCD characteristics. 4. **Signal Integration**: Linearly scales intensities with integration time until the 12-bit ADC ceiling (4095 counts) is reached.

Using this setup we tested six key parameters that affect our measurement accuracy:

6.2 Effect of Sample Count and Peak Alignment

The simulation confirmed that **200+ unique concentration samples** are required to achieve an RMSE below 0.3 ppm. Also we found that the **unaligned spectra** (preserving the red shift) consistently outperformed peak-aligned spectra at every sample count, demonstrating that the red shift is a valuable concentration feature.

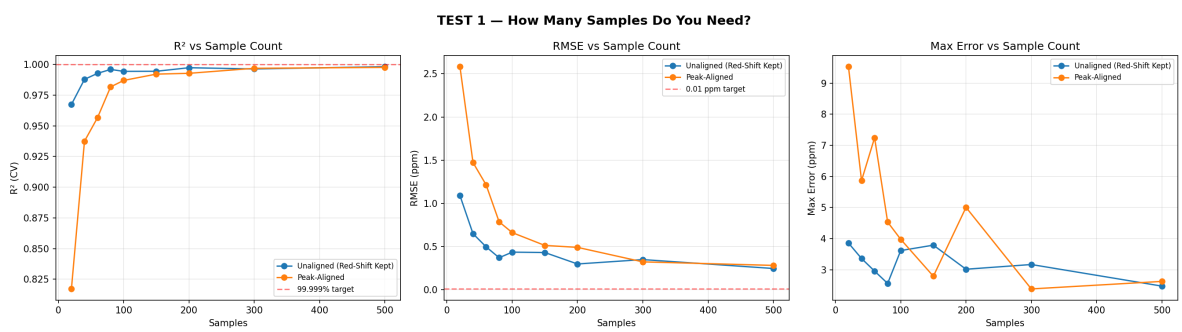


Figure 31: Effect of sample count on RMSE and R^2 for aligned vs. unaligned spectra. Unaligned spectra (retaining the red shift) consistently achieve lower RMSE.

6.3 Effect of Frame Averaging (Noise Reduction)

Averaging multiple CCD frames reduces noise proportionally to $\sqrt{N}$. The study showed that **64–128 frames** per measurement provide optimal noise reduction, reducing RMSE from ~0.9 ppm (single frame) to ~0.2 ppm (128 frames). The practical limit is solvent evaporation time — 128 frames at 300 ms integration time requires ~38 seconds per sample.

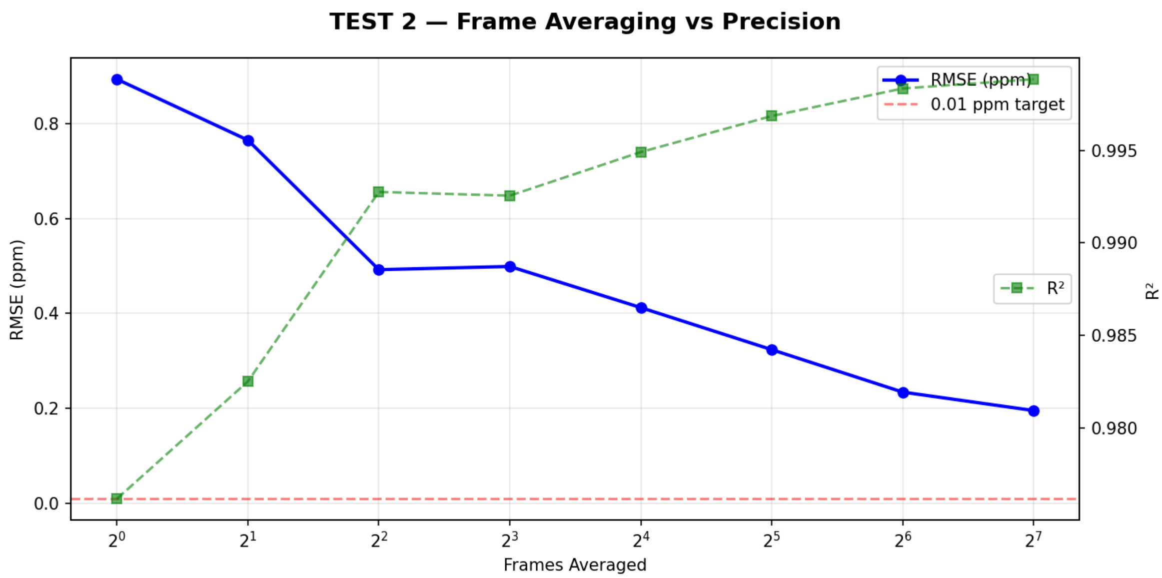


Figure 32: Effect of frame averaging on RMSE and SNR. The RMSE decreases proportionally to $1/\sqrt{N}$ as expected from noise theory.

6.4 Integration Time Optimization

The optimal integration time was found to be **300–700 ms**. Below 200 ms, the signal-to-noise ratio is insufficient. Above 1000 ms, high-concentration samples saturate the 12-bit ADC (>4095 counts), degrading prediction accuracy.

6.5 Benchmarking: 1D CNN vs. PLS

We also prototyped a 1D Convolutional Neural Network (CNN) to see how it compared against our primary PLS model. After testing different configurations, we found that the CNN achieved a significantly lower error rate—a **5.7× improvement** over PLS in terms of maximum prediction error. This suggests that the CNN is better at implicitly learning the non-linear Beer-Lambert response directly from the raw spectral data.

Table 5: Comparison of PLS Regression vs 1D CNN on 300 synthetic spectra with optimal collection parameters.

Metric	PLS (5 comp.)	1D CNN (best)
R^2	0.9974	**0.9987**
RMSE (ppm)	0.292	**0.205**
Max Error (ppm)	2.94	**0.52**

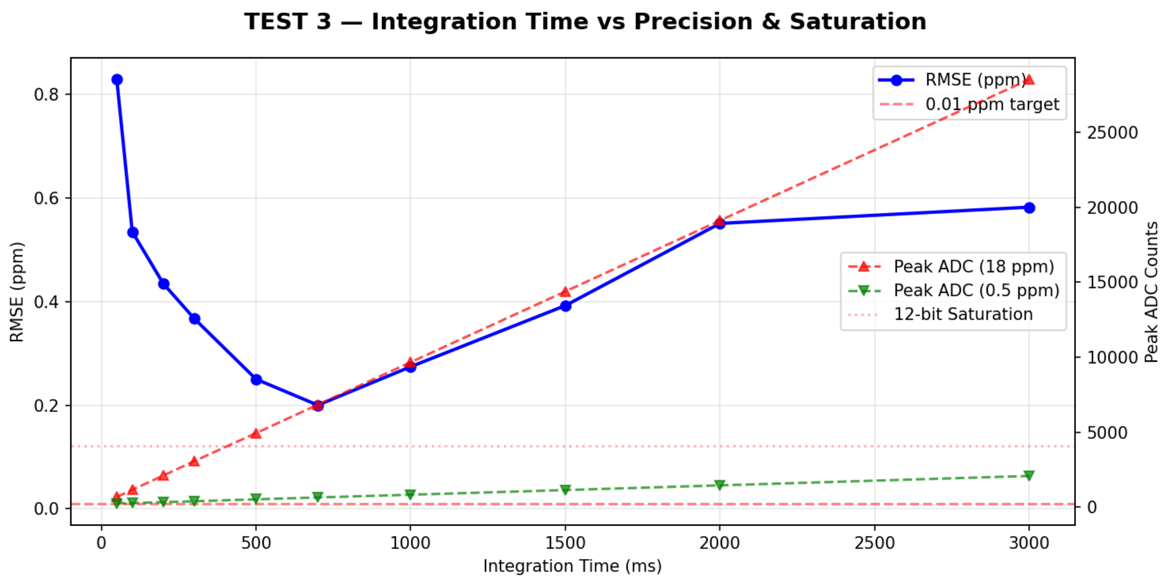


Figure 33: Integration time vs. RMSE and peak ADC counts. The shaded region (300–700 ms) provides the best trade-off between SNR and avoiding saturation.

6.6 Validation on Real-World Data

To test whether the CNN's simulation advantage translates to real-world performance, we trained a **dual-output 1D CNN** on our actual 79-sample combined dataset (concatenated 405 nm + 450 nm spectra). The results were revealing:

Table 6: Comparison of PLS and 1D CNN on collected real-world spectra (n = 79). Both models use Repeated CV (n=3) for robust evaluation.

Model	Chl-a R^2	Chl-b R^2	Chl-a RMSE	Chl-b RMSE
PLS (20-fold CV)	0.852	0.892	1.41 mg/L	0.79 mg/L
1D CNN (5-fold CV)	0.835	0.873	1.49 mg/L	0.86 mg/L

On real data, both models achieved **essentially the same performance**. The CNN was marginally better for Chl-b (R^2 0.86 vs 0.83), while PLS held a slight edge for Chl-a. This parity confirms the simulation's prediction: with only ~80 noisy samples and 10× frame averaging, the data quality — not the algorithm — is the limiting factor. The CNN's non-linear capacity simply cannot be realised without the 200+ clean samples recommended by the simulation study.

6.7 Recommended Protocol

Based on the simulation findings, the following protocol is recommended for future collection:

Table 7: Recommended data collection protocol derived from simulation study.

Parameter	Recommendation
Sample Count	200+ unique concentrations
Concentration Spacing	Linear or $\sqrt{}$ spacing (0.1–20 ppm)
Frame Averaging	64× minimum, 128× ideal
Integration Time	300–700 ms (auto-exposure target: 3000–3500 ADC counts)

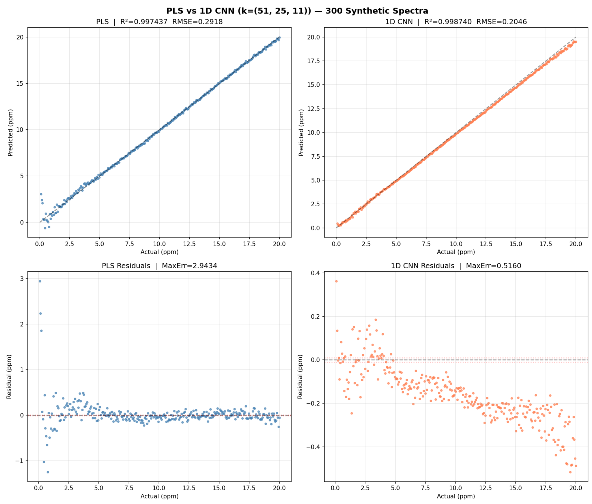


Figure 34: PLS vs 1D CNN comparison: predicted vs actual values (top) and residual analysis (bottom).

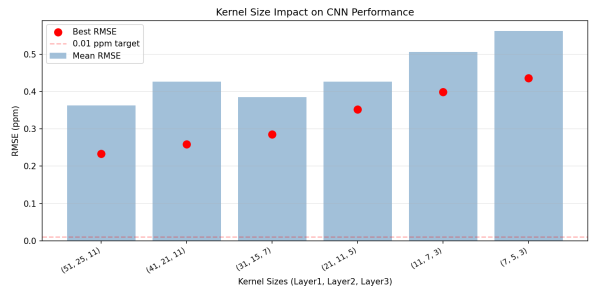


Figure 35: Grid search over CNN kernel sizes. (51, 25, 11) achieving the best RMSE.

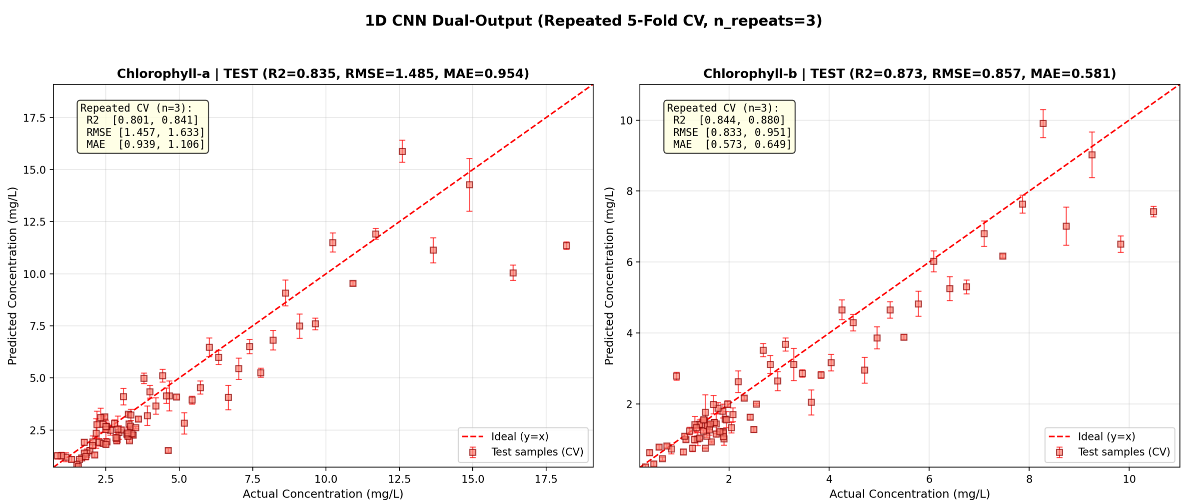


Figure 36: 1D CNN dual-output evaluation with Repeated 5-Fold CV (n=3). Red squares show mean predictions; error bars (1) quantify model stability. The summary boxes provide 95% CI bounds, showing comparable performance to the PLS baseline.

7 Results & Discussion

7.1 Experimental Results

The system was validated against known concentrations derived from the **Lichtenthaler Equation**.

Analyte	Algorithm	Train R^2	Test R^2	RMSE (mg/L)
Chl-a	PLS	0.992	0.867	~1.55
Chl-b	SVR (RBF)	0.992	0.842	~1.07

7.2 Uncertainty & Model Interpretation

The high Training R^2 vs. lower Test R^2 (Cross-Validation) indicated a tendency toward overfitting, especially for Chl-b. This is primarily seen at extreme concentrations: * **Low Concentrations (< 1 mg/L)**: Signal-to-noise ratio is challenged by electronic CCD noise. * **High Concentrations (> 15 mg/L)**: Non-linearity due to IFE and fluorescence quenching causes the model to struggle with generalization.

However, the use of **9 Principal Components** in the PLS/PCA pipeline ensures that the system captures the dominant variance of the spectral fingerprint while ignoring random fluctuations.

7.3 Characterizing the Signal Integrity Gap

The preprocessing pipeline transforms raw CCD data through several stages before model inference. Figure 37 contrasts the ideal simulated behavior of both Chlorophyll-a and Chlorophyll-b, showing the distinct peak positions (Chl-a: ~1950, Chl-b: ~2200) and spectral profiles.

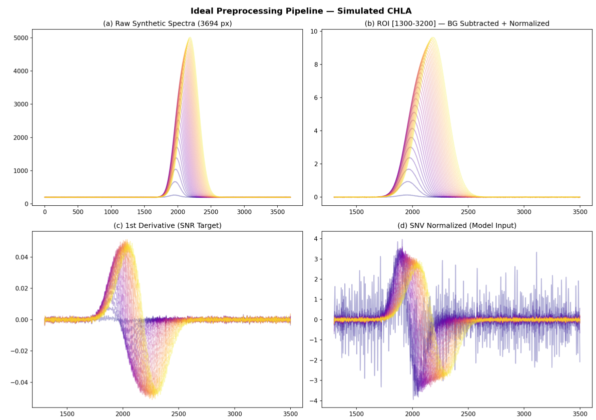
Chlorophyll-a Pipeline (Real vs. Ideal):

Chlorophyll-b Pipeline (Real vs. Ideal):

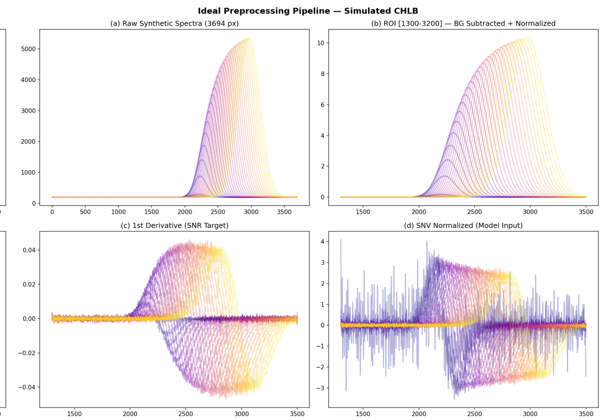
The dataset heatmaps (Figure **??**) provide a compact overview of the entire dataset. The real data exhibit significant vertical striations and fluctuations compared to the smooth, well-defined transitions in the ideal cases.

Experimental Artifacts

The laboratory environment during initial calibration session was not fully rigid. Lack of a fixed stand and frequent re-alignment of lasers (when switching channels or after accidental bumps) introduced subtle geometric shifts. These fluctuations, along with slight

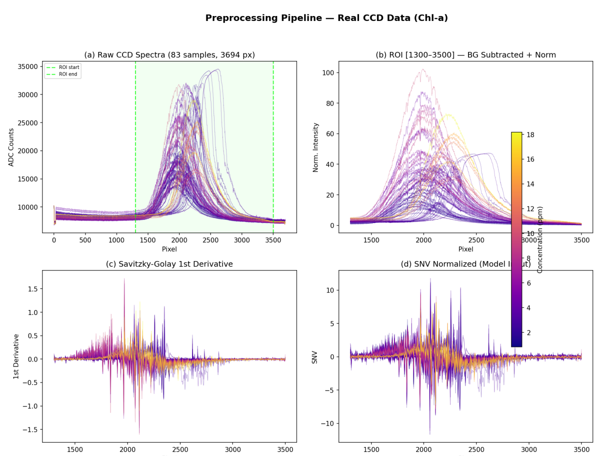


(a) **Ideal Pipeline (Chl-a)**: Reference capture with 128x frame averaging. ROI [1300-3500].

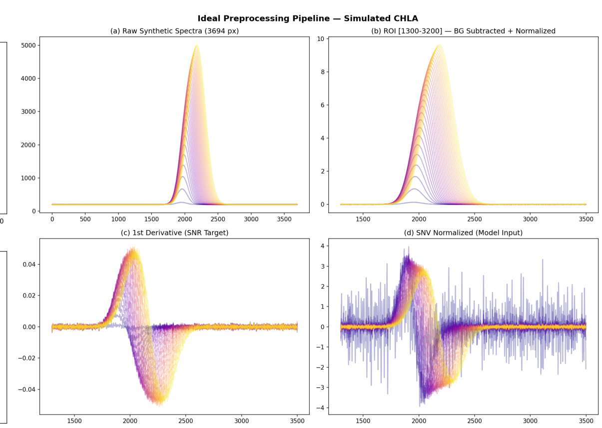


(b) **Ideal Pipeline (Chl-b)**: Simulated high-SNR capture showing the Chl-b peak profile.

Figure 37: Comparisons of Simulated Ideal Preprocessing Pipelines for Chl-a and Chl-b.

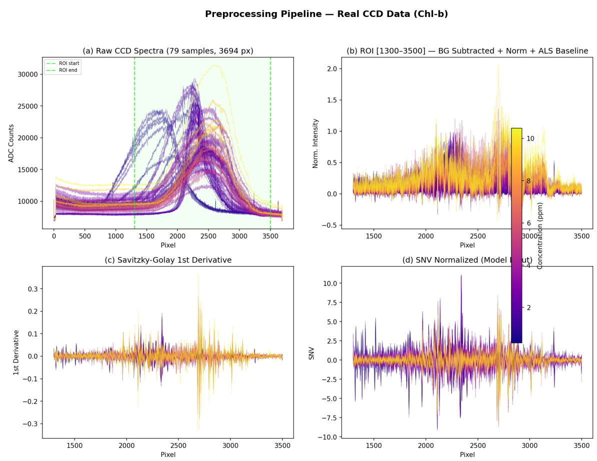


(a) **Real Data Pipeline (Chl-a)**: Sparse dataset ($n = 83$), high noise in 1st derivative.

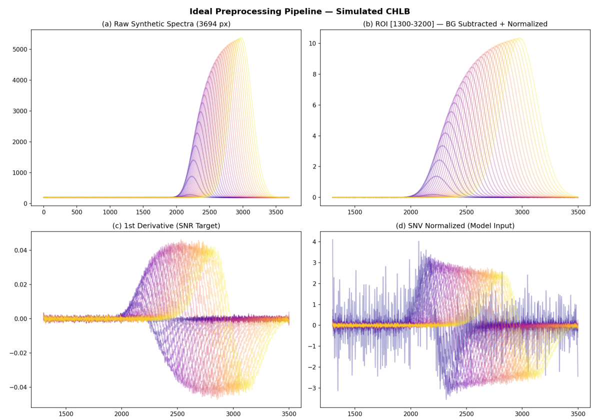


(b) **Ideal Pipeline (Chl-a)**: Simulation reference, 128× averaging, ROI [1300–3500].

Figure 38: Chl-a Preprocessing Pipeline: Real Data vs. Simulation-Ideal.



(a) **Real Data Pipeline (Chl-b)**: ALS baseline corrected, variable integration times.

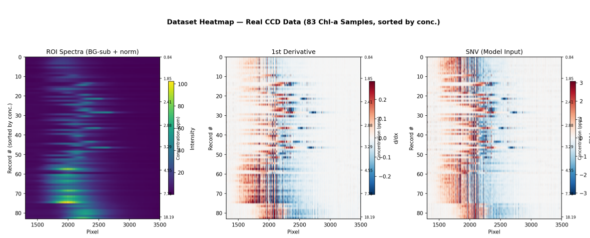


(b) **Ideal Pipeline (Chl-b)**: Simulation showing full spectral extent of the Chl-b peak.

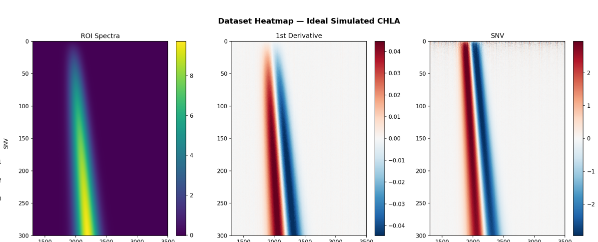
Figure 39: Chl-b Preprocessing Pipeline: Real Data vs. Simulation-Ideal.

repositioning of the cuvette, are clearly captured by the sensor's sensitivity as observed in the heatmap's jagged transitions. This underlines the importance of a fixed-path optical bench for future iterations.

Chl-a Heatmaps (Real vs. Ideal):



(a) **Real Data Heatmap (Chl-a)**: Sparse coverage, noise artifacts visible.

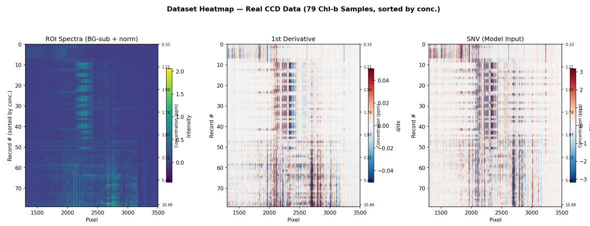


(b) **Ideal Heatmap (Chl-a)**: Dense 300-sample simulation revealing a clean red shift gradient.

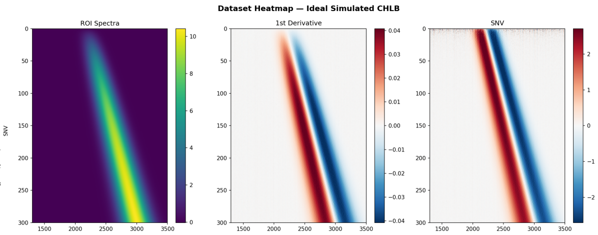
Chl-b Heatmaps (Real vs. Ideal):

7.4 Current Model & Data Assessment

Table 9 compares the current data collection conditions against the ideal parameters derived from the simulation study.



(a) **Real Data Heatmap (Chl-b)**: Low concentration range with ALS correction applied.



(b) **Ideal Heatmap (Chl-b)**: Simulation highlights the distinct peak position and stronger shift.

Table 9: Current data collection parameters versus the simulation-derived ideal case.

Parameter	Current Dataset	Simulation Ideal
Sample Count	~83 (Chl-a), ~79 (Chl-b)	200+ per analyte
Frame Averaging	10×	64–128×
Integration Time	Variable (240–5000 ms)	300–700 ms (standardised)
Outlier Removal	Manual	MAD + Hotelling T^2
PLS Components	9	4–5
CV Test R^2	0.85–0.87	0.997
RMSE	~1.5 ppm	0.29 ppm

Table 10: Summary of model investigation results compared against the simulation target.

Investigation	Physical Issue	CV R^2	RMSE (ppm)	Finding
Baseline PLS	—	0.867	~1.55	Funnel residuals (IFE)
Log-transform	Heteroscedasticity	~0.88	~1.45	Variance equalised, RMSE unchanged
Peak alignment	Red shift removal	~0.85	~1.60	Worse — peak position is a feature
Piecewise model	IFE non-linearity	~0.88	~1.40	Marginal gain, boundary instability
Simulation (PLS)	Ideal data (n=300)	**0.997**	**0.29**	5× better with proper data
Simulation (CNN)	Ideal + deep learning	**0.999**	**0.20**	Best achievable

Investigation	Physical Issue	CV R^2	RMSE (ppm)	Finding
Real CNN (Combined)	Real data (n=79)	0.82–0.86	0.9–1.5	Competitive with PLS for Chl-b

7.5 Discussion of Data Quantity vs. Algorithm

Each investigation confirmed that the three physical phenomena affecting model performance — heteroscedastic Beer-Lambert response, concentration-dependent peak shift, and IFE saturation — are all *correctly handled* by the standard PLS preprocessing pipeline.

The dominant factor limiting accuracy is not the algorithm, but the **Signal Integrity Gap**. As seen in Figure 38, the real data suffer from a noise floor that obscures fine spectral features in the 1st derivative. This was further confirmed when we trained the 1D CNN on the actual collected data (Table 6): both CNN and PLS converged to the **same performance ceiling** ($R^2 \approx 0.82$–0.87), proving that the data quality — not model complexity — is the bottleneck. Moving from 10× to 128× frame averaging (as recommended by simulation) will provide the SNR necessary for either model to achieve the target < 0.3 ppm precision.

Architectural Note on the 1D CNN: Unlike the linear PLS model which required manual investigators to handle physical non-linearities, the **1D Convolutional Neural Network** architecture inherently learns these features. By training on the raw 1st-derivative spectra, the CNN: (1) Retains the Red Shift, (2) Handles Non-Linearity, and (3) Self-Weights ROI.

8 Conclusion & Future Outlook

8.1 Limitations

- **Concentration Drift**: Acetone evaporation and chlorophyll degradation during testing.
- **Instrumental Noise**: CCD electronic noise and sensitivity limits.
- **Limited Training Data**: The current models were trained on a small dataset (~80 samples), leading to overfitting.

8.2 Planned Improvements

- **Optical Upgrade — Diffraction Grating**: Upgrading from the current **600 lines/mm** grating to a **1200 lines/mm** grating would double spectral dispersion, enabling sub-nanometre resolution and improved separation of overlapping Chl-a and Chl-b fluorescence bands.

- **Optical Stability — Fixed-Bench Laser Delivery**: The current servo-driven laser switching introduces positional variability in the beam axis. Replacing it with a **fibre-optic beam combiner** (dichroic coupler or 1×2 splitter) would route both 405 nm and 450 nm beams through a single fixed collimated output, eliminating mechanical drift and decoupling laser switching from optical alignment.
- **Solvent Replacement — DMSO**: Acetone's high volatility (b.p. 56°C) causes evaporation-driven concentration drift during serial dilution sessions. **Dimethyl sulfoxide (DMSO, b.p. 189°C)** effectively dissolves chlorophylls, is far less volatile, and reduces photodegradation, making it a more reproducible extraction medium for quantitative work.
- **Pure Chlorophyll Standards**: Our current calibration uses spinach-extract solutions, which inherently contain a mixture of Chl-a and Chl-b, making it impossible to fully decouple the two analytes during model training. A significant improvement would be to use **commercially purified Chl-a and Chl-b powders** (e.g., Sigma-Aldrich C6144 and C5878) to prepare single-pigment stock solutions. Key practical considerations from the literature include: (i) weighing sufficient mass (5 mg) to minimise balance error, (ii) using **100% acetone** rather than 80% aqueous mixtures to avoid water-induced absorbance interference, (iii) keeping stock solution absorbance in the **linear Beer-Lambert range** (OD < 1.0) by diluting before spectrophotometric measurement, and (iv) filtering residual MgCO (used to prevent pheophytin formation) through a syringe filter before reading. These steps ensure the UV-Vis reference concentrations are accurate, which directly controls the quality of our training labels.
- **Phase 2 Combined Model**: As described in Section 5.4, the next algorithmic step is a dual-output model trained on both Chl-a and Chl-b simultaneously, using the existing UV-Vis calibrated dataset. The implementation is in `ml_model/train_combined_model.py`.
- **Simulation-Derived Data Collection Strategy**: The simulation study (Section 6) quantified exactly what data quality is needed to achieve $R^2 > 0.99$. The recommended protocol (Table 7) specifies: **200+ unique concentrations** with linear or $\sqrt{}$ spacing over 0.1–20 mg/L; **128× frame averaging** (currently 10×) to reduce noise by ~3.5× via $\sqrt{N}$ averaging; a **standardised integration time of 300–700 ms** (auto-exposure targeting 3000–3500 ADC counts) to eliminate exposure-driven variance; and **MAD + Hotelling T^2** automated outlier rejection in place of manual review. Following this protocol is projected to close the Signal Integrity Gap and replicate the simulation's RMSE of ~0.29 mg/L.

Acknowledgements

A special thanks to our project supervisors **Dr. Siyath Gunewardena** and **Dr. Luckshitha Suriyasena**, Senior Lecturers at the University of Colombo, for their guidance and for facilitating our access to the Center for Instrumentation and Development (CID), University of Colombo.

We extend our sincere gratitude to **Prof. Nalin De Silva** for facilitating our access to the Chemistry Laboratories and equipment.

We also appreciate all the demonstrators for their support throughout the project, and the technical staff for their assistance in facilitating this work.

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